VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Artificial Intelligence (23CS5PCAIN)

Submitted by
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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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B.M.S. College of Engineering,
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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Artificial Intelligence (23CS5PCAIN)" carried out by **Shreya Jaganatha Gowda(1BM23CS352),** who is a bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Artificial Intelligence (23CS5PCAIN) work prescribed for the said degree.

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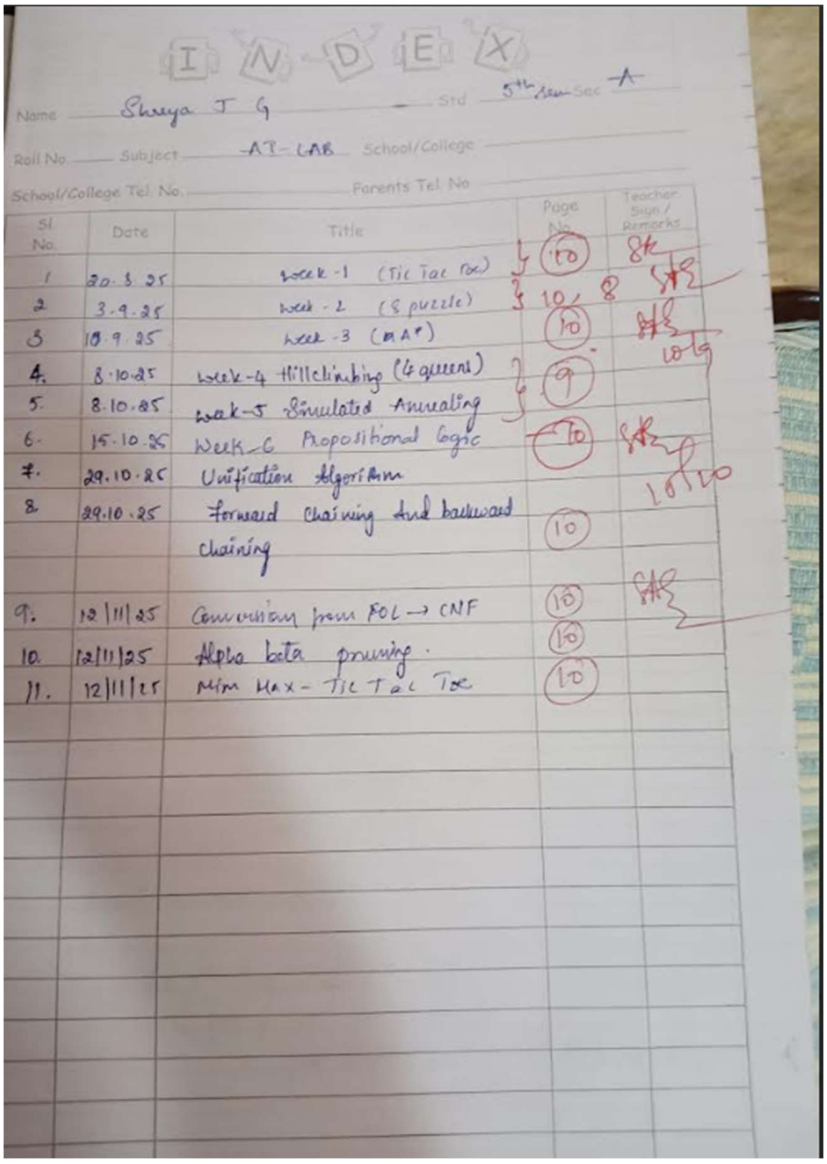
Github Link:

[Shreyajg/AI-LAB]



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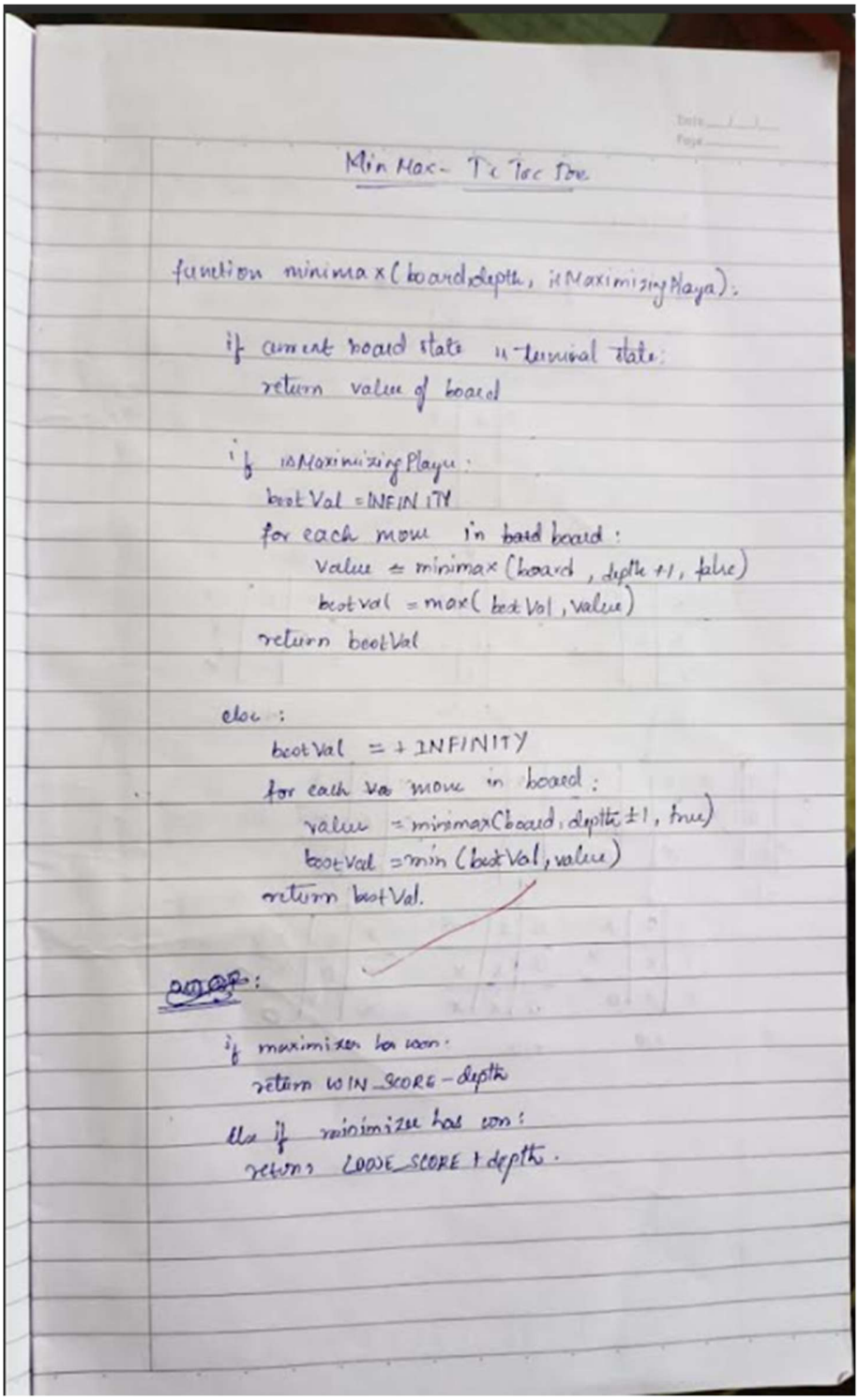
Program 1

Implement Tic –Tac –Toe Game

Algorithm:



Min Max - Tic Tac Toe

function minimax(board, depth, isMaximizingPlaya):

if current board state is terminal state:
return value of board

if isMaximizingPlaya:
bestVal = INFINITY
for each move in bestboard:
value = minimax(board, depth+1, false)
bestval = max(bestVal, value)
return bestVal

else:
bestVal = +INFINITY
for each va move in board:
value = minimax(board, depth+1, true)
bestVal = min(bestVal, value)
return bestVal.

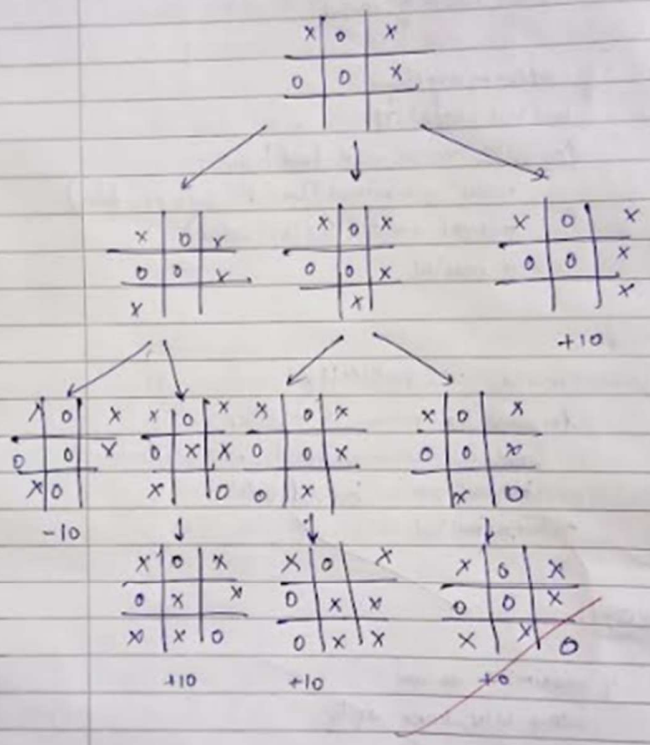
:

if maximizer has won:
return WIN_SCORE - depth

else if minimizer has won:
return LOOSE_SCORE + depth.

OUTPUT:

The value of the best move is : 10
the optimal move is:
ROW: 2 COL: 0

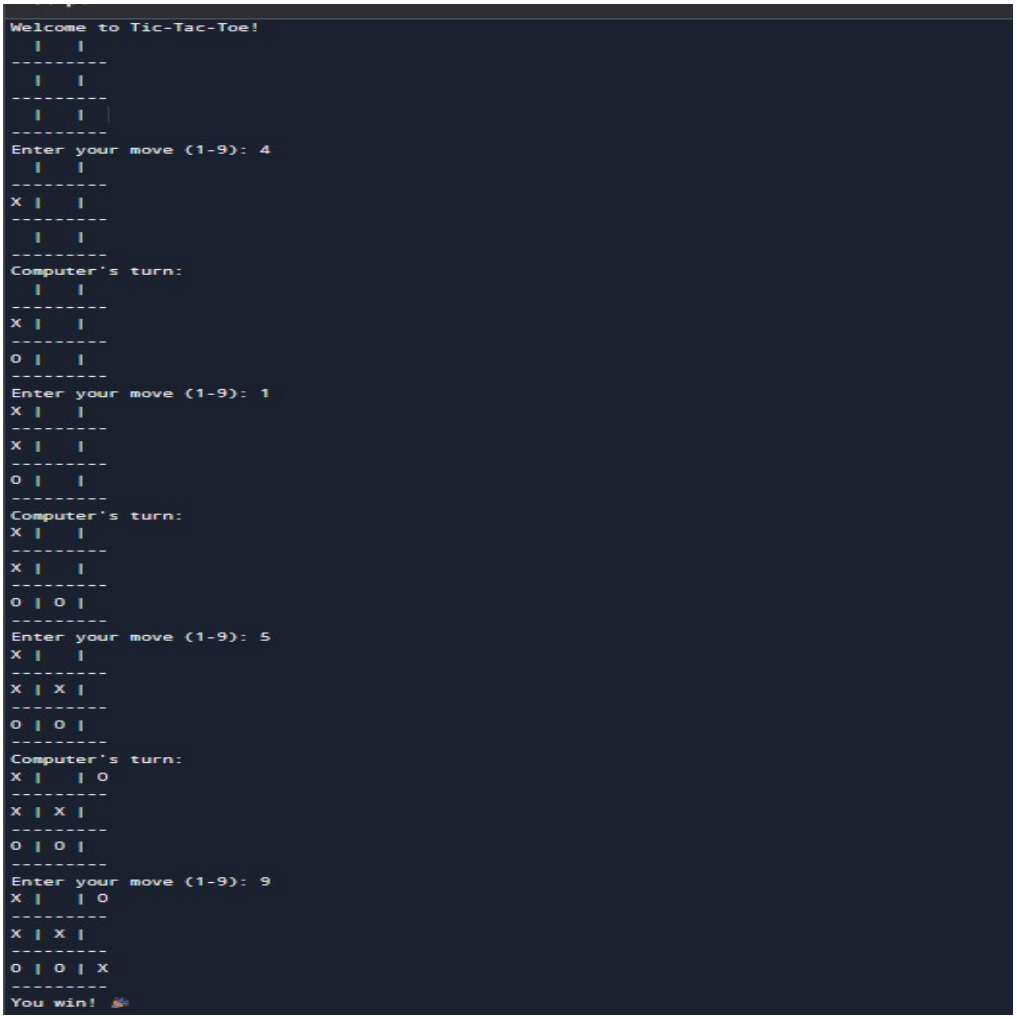


Code:

```
import random
board = [' ' for _ in range(9)]

def print_board():
    print()
    for i in range(3):
        print(" " + " | ".join(board[i*3:(i+1)*3]))
        if i < 2:
            print("---+---+---")
    print()
def check_winner(player):
    win_conditions = [
        [0,1,2], [3,4,5], [6,7,8],
        [0,3,6], [1,4,7], [2,5,8],
        [0,4,8], [2,4,6]
    ]
    for cond in win_conditions:
        if all(board[i] == player for i in cond):
            return True
    return False
def is_full():

return all(cell != ' ' for cell in board) def
player_move():
        while True:
            try:
                move = int(input("Enter your move (1-9): ")) - 1
                if move < 0 or move >= 9:
                    print("Invalid move. Choose between 1-9.")
                elif board[move] != ' ':
                    print("That spot is taken.")
                else:
                    board[move] = 'X'
                    break
            except ValueError:
                print("Please enter a valid number.")
    def ai_move():
        empty_spots = [i for i, val in enumerate(board) if val == ' ']
        move = random.choice(empty_spots)
        board[move] = 'O'
        print(f"System placed 'O' in position {move+1}")
    def play_game():
        print("Welcome to Tic Tac Toe!")
        print_board()
        while True:
            player_move()
```

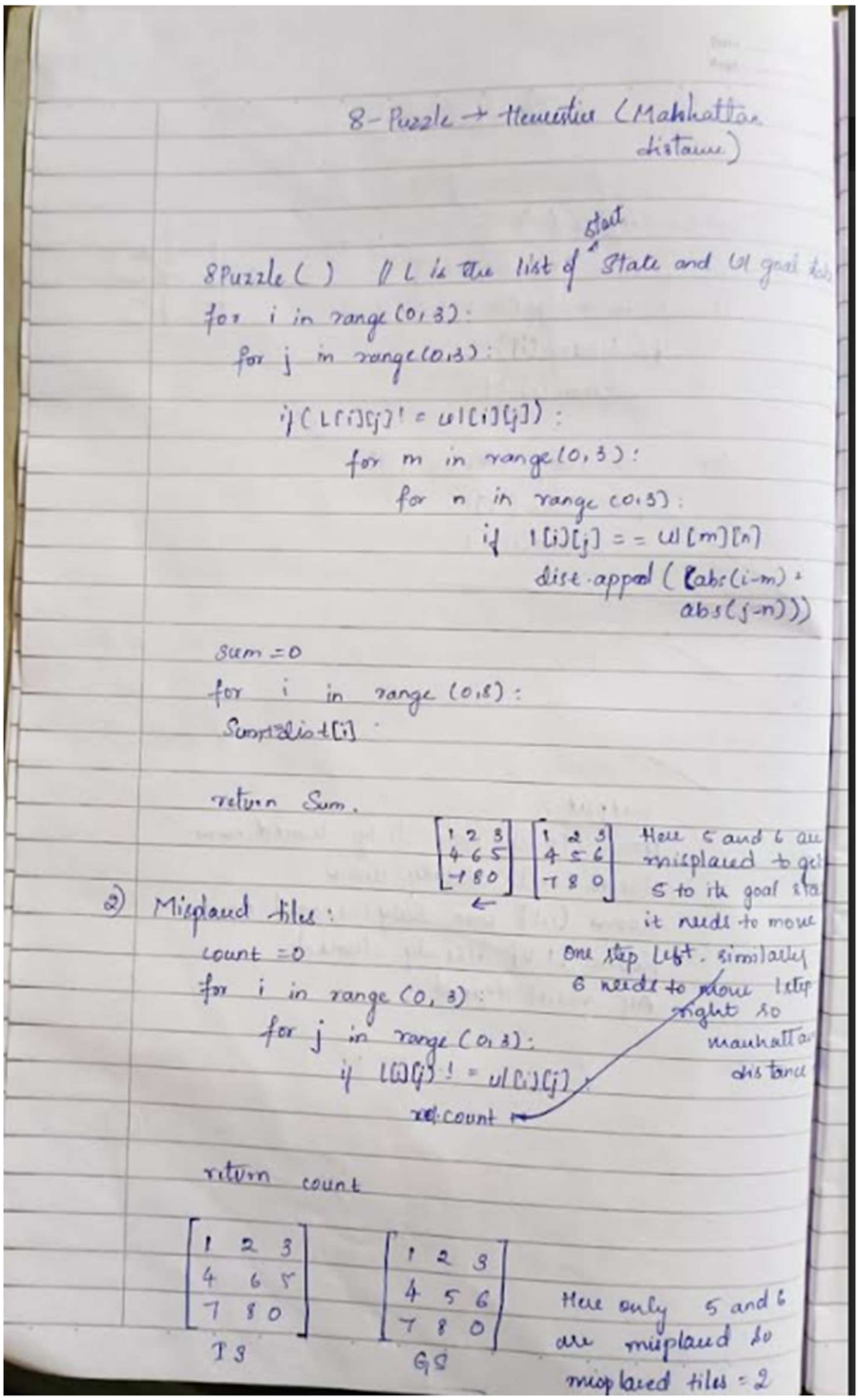
```
        print_board()
        if check_winner('X'):
            print("Congratulations! You win!")
            break
        if is_full():
            print("It's a tie!")
            break
        ai_move()
        print_board()
        if check_winner('O'):
            print("System wins. Better luck next time!")
            break
        if is_full():
            print("It's a tie!")
            break
if __name___== "__main__":
    play_game()
```

ScreenShots:



Program 2:

Solve 8 puzzle problems.
Algorithm:



8-Puzzle → Heuristics (Manhattan distance)

8Puzzle( ) // L is the list of start State and U goal state

```
for i in range(0,3):
    for j in range(0,3):
        if(L[i][j] != U[i][j]):
            for m in range(0,3):
                for n in range(0,3):
                    if I[i][j] == U[m][n]
                        dist.append(abs(i-m)+abs(j-n))
sum=0
for i in range(0,8):
    Sum+=dist[i]
return Sum.
```

Here 5 and 6 are misplaced to get 5 to its goal state it needs to move one step left. similarly 6 needs to move 1 step right so manhattan distance

2) Misplaced tiles:

```
count=0
for i in range(0,3):
    for j in range(0,3):
        if L[i][j] != U[i][j]:
            count
return count
```

Here only 5 and 6 are misplaced so misplaced tiles = 2

IS GS

Output Manhattan Distance :

enter Initial state
1
2
3
4
6
5
7
8

Manhattan distance = 2

1	2	3
4	6	5
7	8	0

Initial state

1	2	3
4	5	6
7	8	0

Goal state

Manhattan distance = $\sum$ (abs(goalposrow - posrow) + (goalposcol - poscol))

Here only 6 and 5 are misplaced

0 + 0 + 0 + 0 + 1 + 1 + 0 + 0 + 0 = 2//

Misplaced tiles : 8 - 3 = 5

2 as there are 2 misplaced tiles.

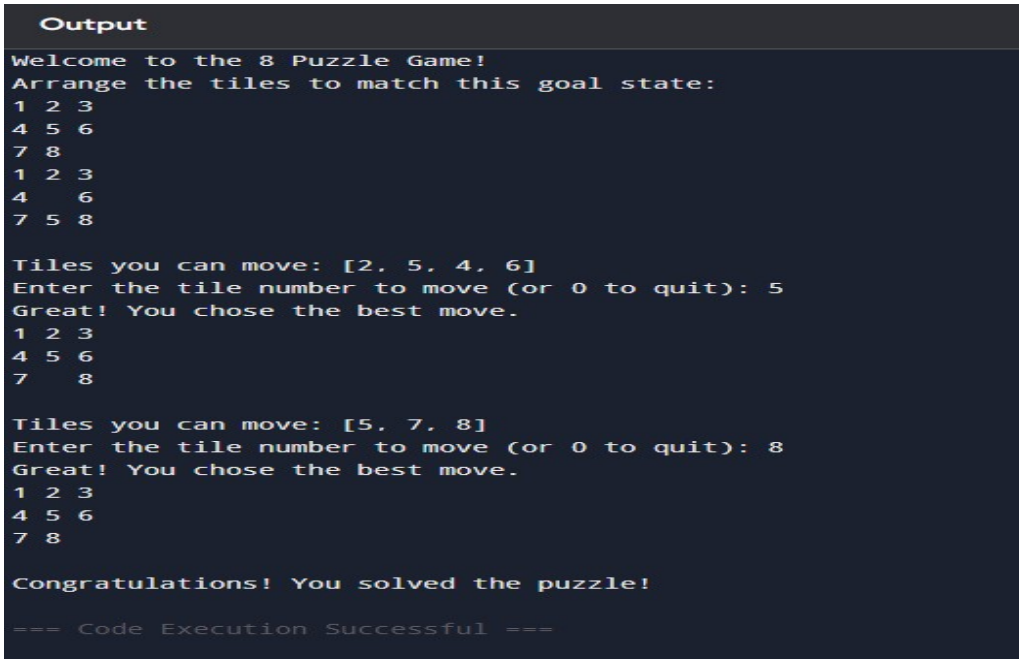
Code:

```
import copy
def print_board(board):
    for row in board:
        print(' '.join(str(x) if x != 0 else ' ' for x in row))
    print()
def find_zero(board):
    for i in range(3):
        for j in range(3):
            if board[i][j] == 0:
                return i, j
def is_solved(board):
    solved = [1,2,3,4,5,6,7,8,0]
    flat = [num for row in board for num in row]

return flat == solved
def valid_moves(zero_pos):
    i, j = zero_pos
    moves = []
    if i > 0: moves.append((i-1, j))
    if i < 2: moves.append((i+1, j))
    if j > 0: moves.append((i, j-1))
    if j < 2: moves.append((i, j+1))
    return moves
def correct_tiles_count(board):
    """Count how many tiles are in their correct position."""
    count = 0
    goal = [1,2,3,4,5,6,7,8,0]
    flat = [num for row in board for num in row]
    for i in range(9):
        if flat[i] != 0 and flat[i] == goal[i]:
            count += 1
    return count
def get_user_move(board):
    zero_pos = find_zero(board)
    moves = valid_moves(zero_pos)
    movable_tiles = [board[i][j] for (i,j) in moves]
    print(f"Tiles you can move: {movable_tiles}")
    while True:
        try:
            move = int(input("Enter the tile number to move (or 0 to quit): "))
            if move == 0:
                return None
            if move in movable_tiles:
                return move
            else:
                print("Invalid tile. Please choose a tile adjacent to the empty space.")
```

```
            except ValueError:
                print("Please enter a valid number.")
        def evaluate_move(board, tile):
            """Compare user move to all possible moves and tell if it's best/worst."""
            zero_pos = find_zero(board)
            moves = valid_moves(zero_pos)
            movable_tiles = [board[i][j] for (i,j) in moves]
            scores = {}
            for t in movable_tiles:
                temp_board = copy.deepcopy(board)
                make_move(temp_board, t)
                scores[t] = correct_tiles_count(temp_board)
            user_score = scores[tile]
            best_score = max(scores.values())
            worst_score = min(scores.values())

if user_score == best_score and user_score == worst_score: return
"Your move is the only possible move."
            elif user_score == best_score:
                return "Great! You chose the best move."
            elif user_score == worst_score:
                return "Oops! You chose the worst move."
            else:
                return "Your move is neither the best nor the worst."
        def make_move(board, tile):
            zero_i, zero_j = find_zero(board)
            for i, j in valid_moves((zero_i, zero_j)):
                if board[i][j] == tile:
                    board[zero_i][zero_j], board[i][j] = board[i][j], board[zero_i][zero_j]
                    return
        def main():
            board = [
                [1, 2, 3],
                [4, 0, 6],
                [7, 5, 8]
            ]
            print("Welcome to the 8 Puzzle Game!")
            print("Arrange the tiles to match this goal state:")
            print("1 2 3\n4 5 6\n7 8  ")
            while True:
                print_board(board)
                if is_solved(board):
                    print("Congratulations! You solved the puzzle!")
                    break
                move = get_user_move(board)
                if move is None:
                    print("Game exited. Goodbye!")
```

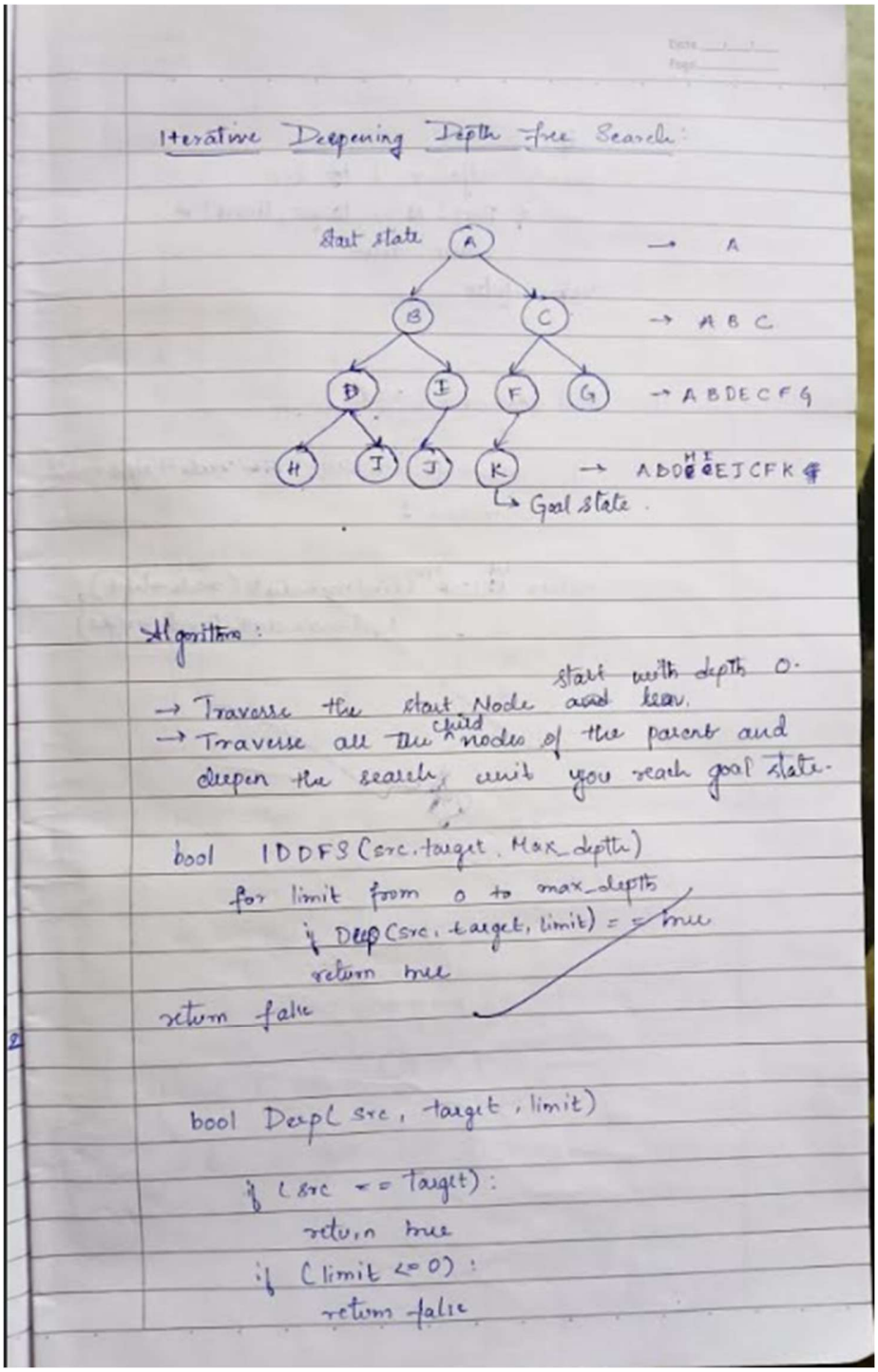
```
            break
        feedback = evaluate_move(board, move)
        print(feedback)
        make_move(board, move)
if __name__ == "__main__":
    main()
```

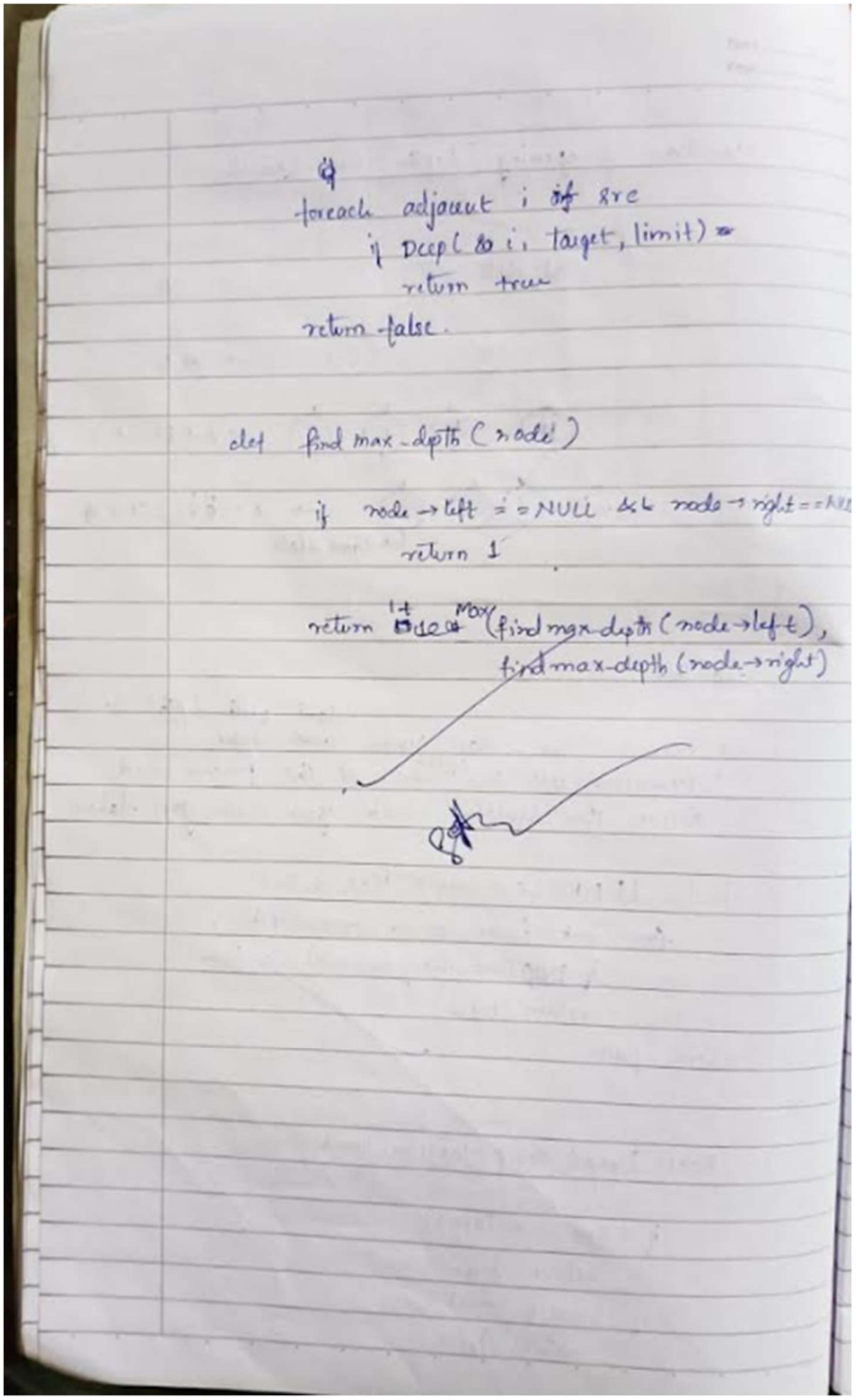
ScreenShot:



Program 3:

Implement Iterative deepening search algorithm.
Algorithm:





```
foreach adjacent i of src
    if Dcpl(& i, target, limit) =
        return true
return false.

def find max_depth (node)

    if node -> left == NULL && node -> right == NU
        return 1

return 1 + max(find max_depth (node->left),
               find max_depth (node->right)
```

Code:

```
import copy
def get_puzzle(name):
    print(f"\nEnter the {name} puzzle (3x3, use -1 for blank):")
    puzzle = []
    for i in range(3):
        row = list(map(int, input(f"Row {i+1} (space-separated 3 numbers): ").split()))
        puzzle.append(row)
    return puzzle
def move(temp, movement):
    for i in range(3):
        for j in range(3):
            if temp[i][j] == -1:
            if movement == "up" and i > 0:
                temp[i][j], temp[i-1][j] = temp[i-1][j], temp[i][j]
            elif movement == "down" and i < 2:
                temp[i][j], temp[i+1][j] = temp[i+1][j], temp[i][j]
            elif movement == "left" and j > 0:
                temp[i][j], temp[i][j-1] = temp[i][j-1], temp[i][j]
            elif movement == "right" and j < 2:
                temp[i][j], temp[i][j+1] = temp[i][j+1], temp[i][j]
            return temp
    return temp
def dls(puzzle, depth, limit, last_move, goal):
    if puzzle == goal:
        return True, [puzzle], []
    if depth >= limit:
        return False, [], []
    for move_dir, opposite in [("up","down"), ("left","right"), ("down","up"), ("right","left")]:
        if last_move == opposite:  # avoid direct backtracking
            continue
        temp = copy.deepcopy(puzzle)
        new_state = move(temp, move_dir)
        if new_state != puzzle:  # valid move
            found, path, moves = dls(new_state, depth+1, limit, move_dir, goal)
            if found:
                return True, [puzzle] + path, [move_dir] + moves
    return False, [], []
def ids(start, goal):
    for limit in range(1, 50):  # reasonable max depth
        print(f"\nTrying depth limit = {limit}")
        found, path, moves = dls(start, 0, limit, None, goal)
        if found:
            print("Solution found!")
            for step in path:
                print(step)
            print("Moves:", moves)
```

```
            print("Path cost =", len(path)-1)
            return
    print(" Solution not found within depth limit.")
start_puzzle = get_puzzle("start")
goal_puzzle = get_puzzle("goal")
print("\n~~~~~~~~~~~~~ IDDFS ~~~~~~~~~~~~~")
ids(start_puzzle, goal_puzzle)
```

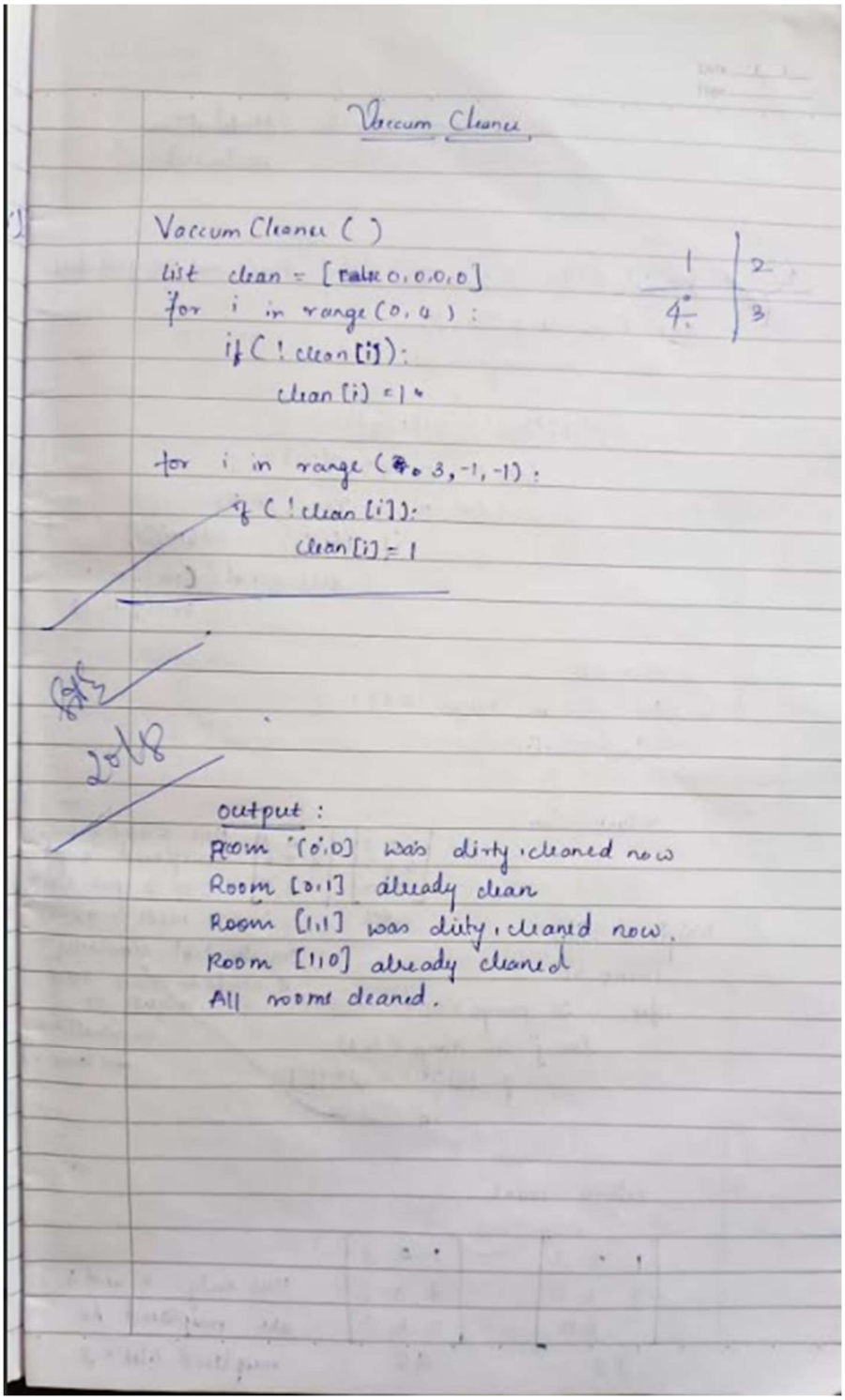
ScreenShot:

Output

```
Enter the start puzzle (3x3, use -1 for blank):
Row 1 (space-separated 3 numbers): 1 2 3
Row 2 (space-separated 3 numbers): 4 7 8
Row 3 (space-separated 3 numbers): 5 6 -1

Enter the goal puzzle (3x3, use -1 for blank):
Row 1 (space-separated 3 numbers): 1 2 3
Row 2 (space-separated 3 numbers): 4 5 6
Row 3 (space-separated 3 numbers): 7 8 -1

~~~~~~~~~~~~~ IDDFS ~~~~~~~~~~~~~

Trying depth limit = 1

Trying depth limit = 2

Trying depth limit = 3

Trying depth limit = 4

Trying depth limit = 5

Trying depth limit = 6

Trying depth limit = 7

Trying depth limit = 8

Trying depth limit = 9

Trying depth limit = 10
```

Program 4:

Implement a vacuum cleaner agent.
Algorithm:



Vaccum Cleaner

```
Vaccum Cleaner ( )
list clean = [false 0,0,0,0]
for i in range(0, 4):
    if(!clean[i]):
        clean[i] = 1

for i in range(3, -1, -1):
    if(!clean[i]):
        clean[i] = 1
```

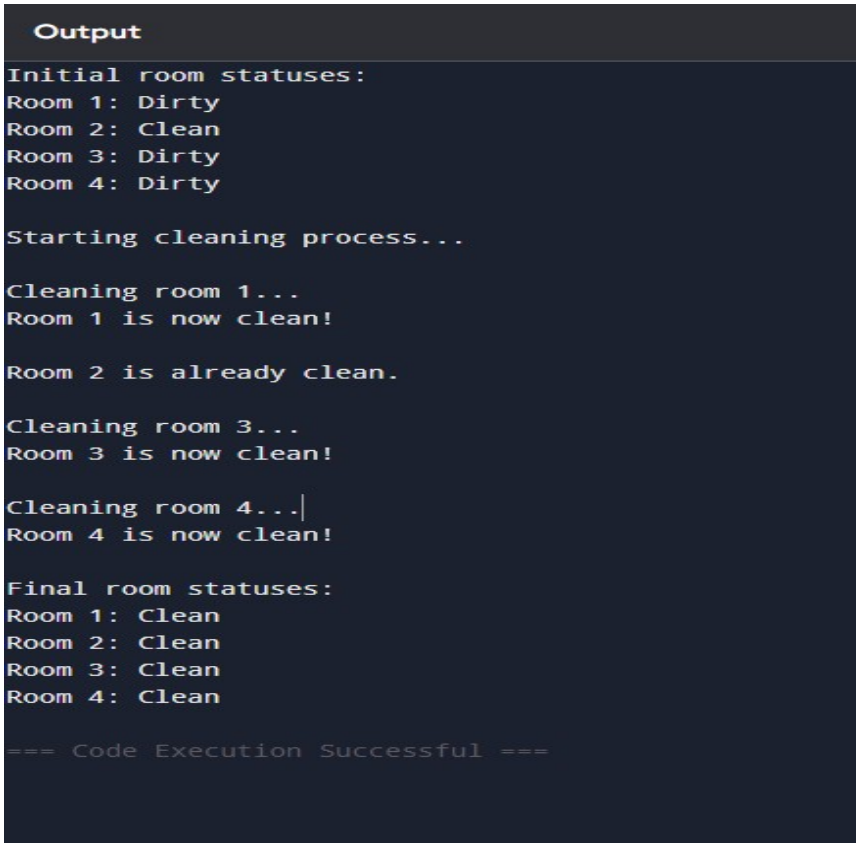
1 | 2
4 | 3

output :
Room [0,0] was dirty, cleaned now
Room [0,1] already clean
Room [1,1] was dirty, cleaned now.
Room [1,0] already cleaned
All rooms cleaned.

Code:

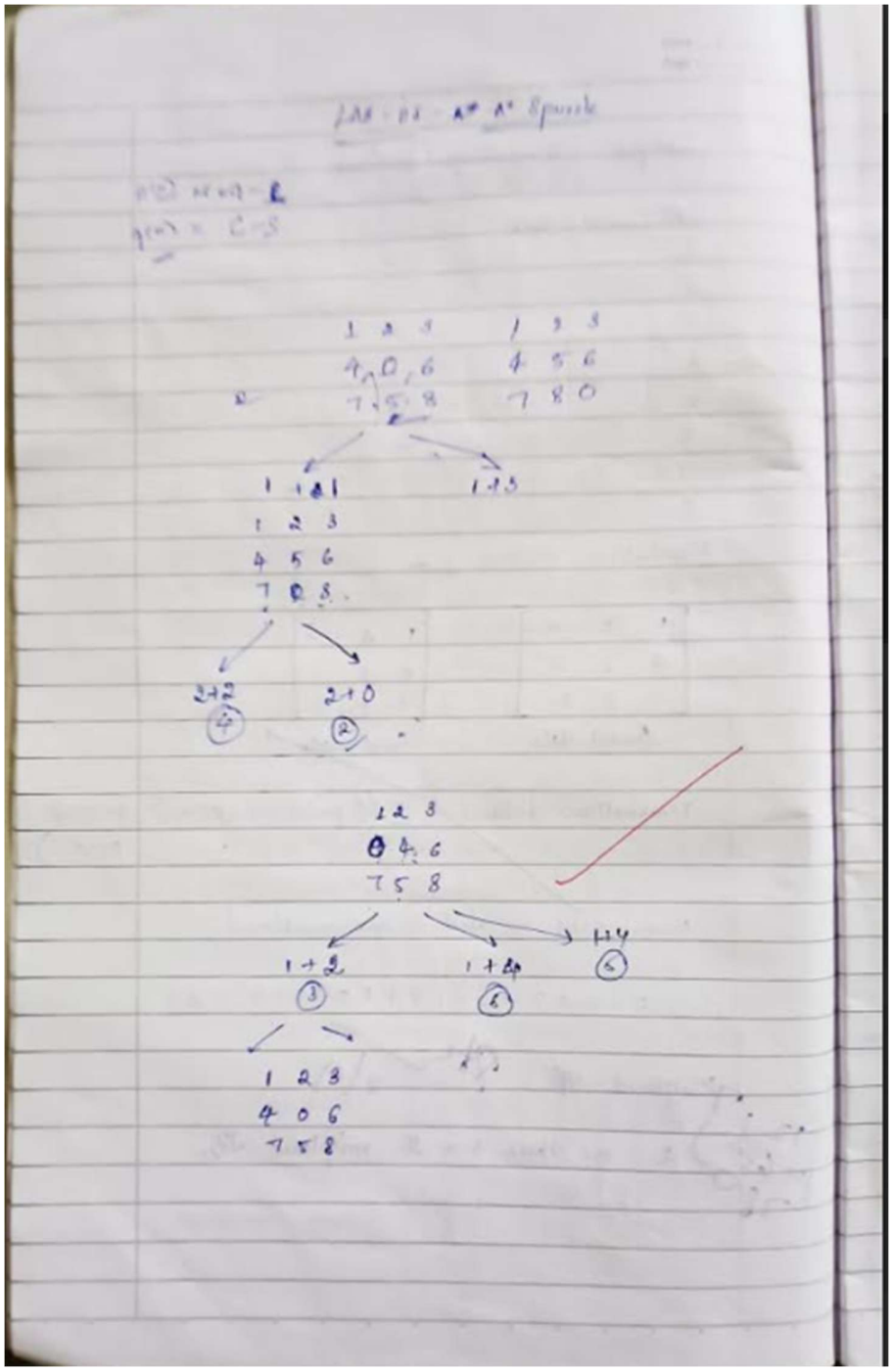
```
def show_rooms_status(rooms):
    for room_number, status in rooms.items():
        print(f"Room {room_number}: {'Clean' if status else 'Dirty'}")
def clean_room(rooms, room_number):
    if rooms[room_number]:
        print(f"Room {room_number} is already clean.")
    else:
        print(f"Cleaning room {room_number}...")
        rooms[room_number] = True
        print(f"Room {room_number} is now clean!")
def clean_all_rooms(rooms):
    print("Initial room statuses:")
    show_rooms_status(rooms)

    print("\nStarting cleaning process...\n")
    for room_number in rooms:
        clean_room(rooms, room_number)
        print()
    print("Final room statuses:")
    show_rooms_status(rooms)
if __name__ == "__main__":
    rooms = {
        1: False,
        2: True,
        3: False,
        4: False
    }
    clean_all_rooms(rooms)
```

ScreenShot:


Output
Initial room statuses:
Room 1: Dirty
Room 2: Clean
Room 3: Dirty
Room 4: Dirty
Starting cleaning process...
Cleaning room 1...
Room 1 is now clean!
Room 2 is already clean.
Cleaning room 3...
Room 3 is now clean!
Cleaning room 4...
Room 4 is now clean!
Final room statuses:
Room 1: Clean
Room 2: Clean
Room 3: Clean
Room 4: Clean
=== Code Execution Successful ===

Program 5:

Implement A* search algorithm.
Algorithm:



1 2 3
4 8 _
7 6 5

h(n) = MH.

Puzzle using A* Algorithm

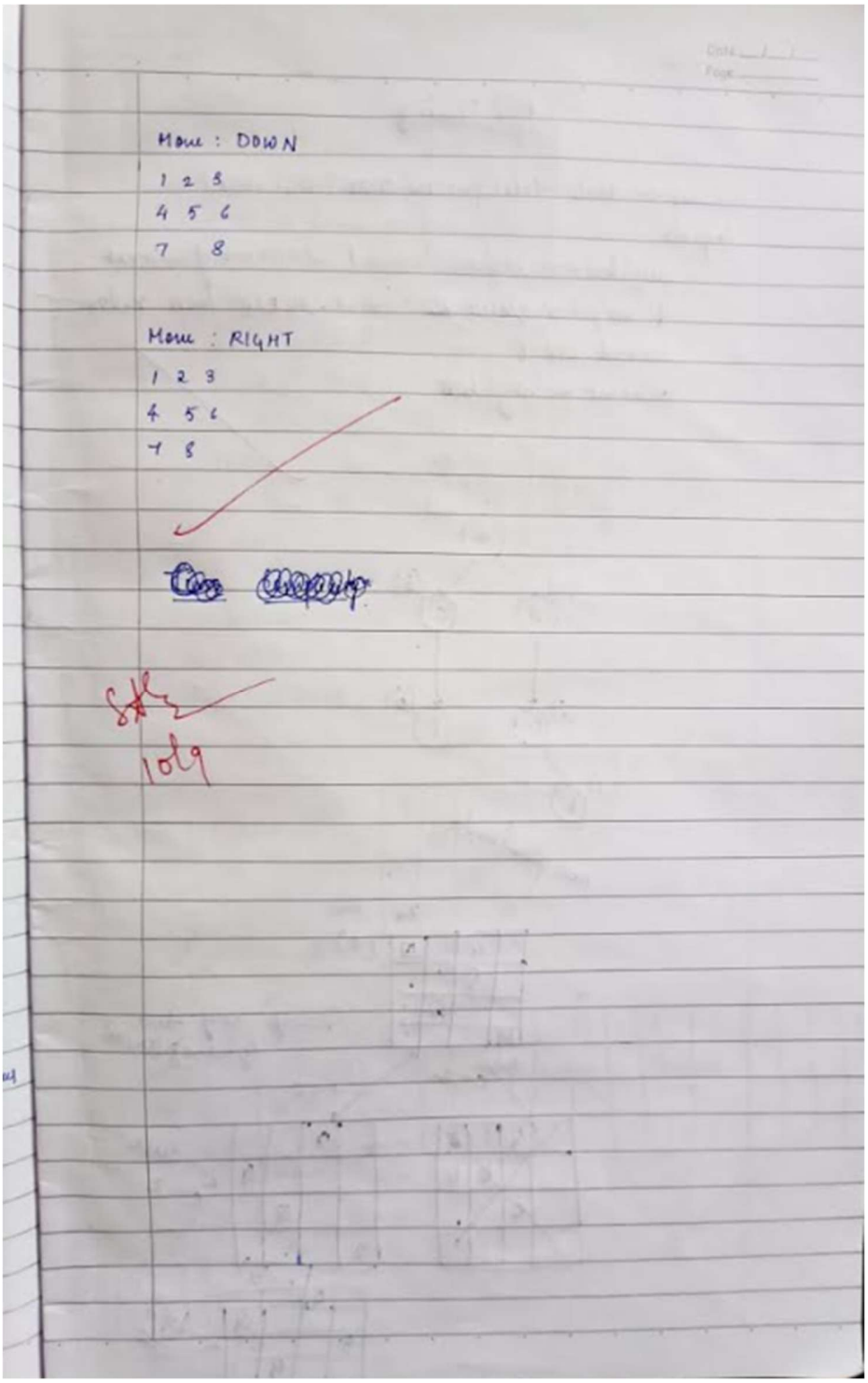
```
A_star_puzzle (start, goal):

    open_list ← {start}
    closed_list ← ϕ
    g[start] = 0
    f[start] = g[start] + h[start]

    while open list is not empty:

        current ← node in open_list with lowest value
        if (current == goal) return

        reconstruct_path(current)
        remove current from open_list
        add current to closed_list
        for each neighbour in get_neighbour(current)
            if neighbour in closed_list
                continue
            tentative_g = g[neighbour] + 1
            f[neighbour] = g[neighbour] + h[neighbour]
            if neighbour not in open_list
                add neighbour to open_list
        return "Soln found"
    return "No Soln found"
```

Move : DOWN

1 2 3
4 5 6
7 8

Move : RIGHT

1 2 3
4 5 6
7 8



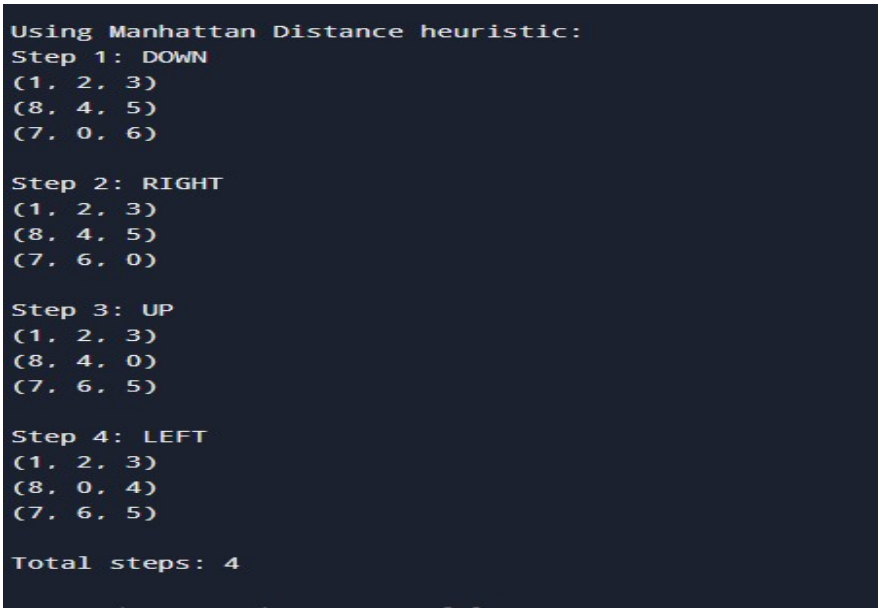
Code:

```
from heapq import heappush, heappop
goal_state = [
    [1, 2, 3],
    [8, 0, 4],
    [7, 6, 5]
]
directions = [(-1, 0), (1, 0), (0, -1), (0, 1)]
direction_names = ["UP", "DOWN", "LEFT", "RIGHT"]
def misplaced_tiles(state):
    count = 0
    for i in range(3):
        for j in range(3):

if state[i][j] != 0 and state[i][j] != goal_state[i][j]: count += 1
    return count
def manhattan_distance(state):
    distance = 0
    for i in range(3):
        for j in range(3):
            tile = state[i][j]
            if tile != 0:
                goal_x, goal_y = divmod(tile - 1, 3)
                distance += abs(i - goal_x) + abs(j - goal_y)
    return distance
def get_neighbors_with_actions(state):
    neighbors = []
    for i in range(3):
        for j in range(3):
            if state[i][j] == 0:
                x, y = i, j
                break
    for (dx, dy), action in zip(directions, direction_names):
        nx, ny = x + dx, y + dy
        if 0 <= nx < 3 and 0 <= ny < 3:
            new_state = [list(row) for row in state]
            new_state[x][y], new_state[nx][ny] = new_state[nx][ny], new_state[x][y]
            neighbors.append((new_state, action))
    return neighbors
def state_to_tuple(state):
    return tuple(tuple(row) for row in state)
def reconstruct_path(came_from, current):
    actions = []
    states = []
    while current in came_from:
        prev_state, action = came_from[current]
```

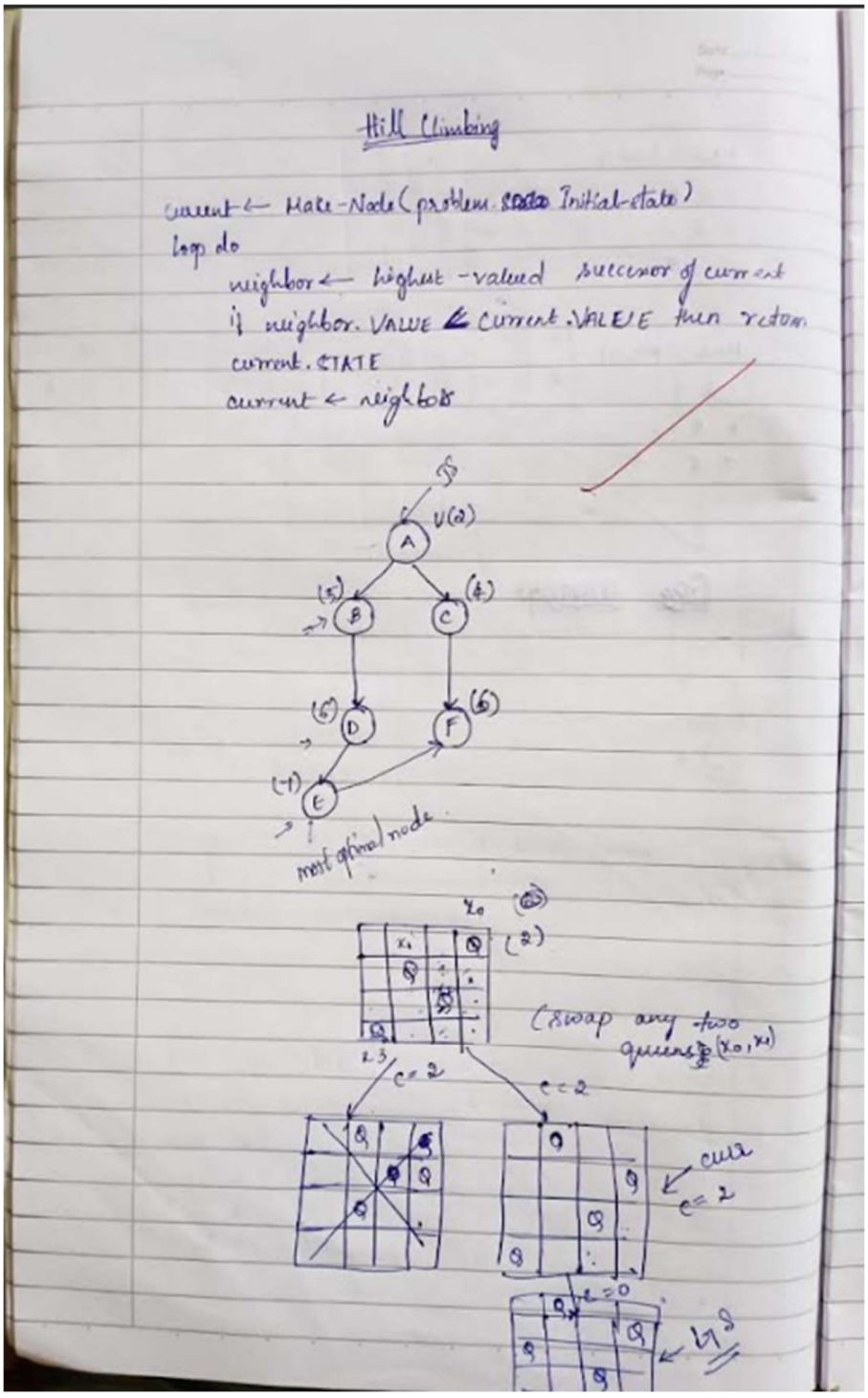
```
            actions.append(action)
            states.append(current)
            current = prev_state
        states.append(current)
        actions.reverse()
        states.reverse()
        return actions, states
    def a_star_search_with_steps(initial_state, heuristic_func):
        open_list = []
        closed_set = set()
        g_score = {state_to_tuple(initial_state): 0}
        f_score = {state_to_tuple(initial_state): heuristic_func(initial_state)}
        came_from = {}
        heappush(open_list, (f_score[state_to_tuple(initial_state)], initial_state))

        while open_list:
            _, current_state = heappop(open_list)
            current_t = state_to_tuple(current_state)
            if current_state == goal_state:
                return reconstruct_path(came_from, current_t)
            closed_set.add(current_t)
            for neighbor, action in get_neighbors_with_actions(current_state):
                neighbor_t = state_to_tuple(neighbor)
                if neighbor_t in closed_set:
                    continue
                tentative_g = g_score[current_t] + 1
                if neighbor_t not in g_score or tentative_g < g_score[neighbor_t]:
                    came_from[neighbor_t] = (current_t, action)
                    g_score[neighbor_t] = tentative_g
                    f_score[neighbor_t] = tentative_g + heuristic_func(neighbor)
                    heappush(open_list, (f_score[neighbor_t], neighbor))
        return None, None
    def print_path(actions, states):
        for i, (action, state) in enumerate(zip(actions, states[1:]), 1):
            print(f"Step {i}: {action}")
            for row in state:
                print(row)
            print()
    initial_state = [
        [1, 2, 3],
        [8, 0, 5],
        [7, 4, 6]
    ]
    print("Using Misplaced Tiles heuristic:")
    actions, states = a_star_search_with_steps(initial_state, misplaced_tiles)
    if actions:
        print_path(actions, states)
```

```
    print("Total steps:", len(actions))
else:
    print("No solution found.")
print("\nUsing Manhattan Distance heuristic:")
actions, states = a_star_search_with_steps(initial_state, manhattan_distance)
if actions:
    print_path(actions, states)
    print("Total steps:", len(actions))
else:
    print("No solution found.")
```

ScreenShot:



b. Implement Hill Climbing Algorithm

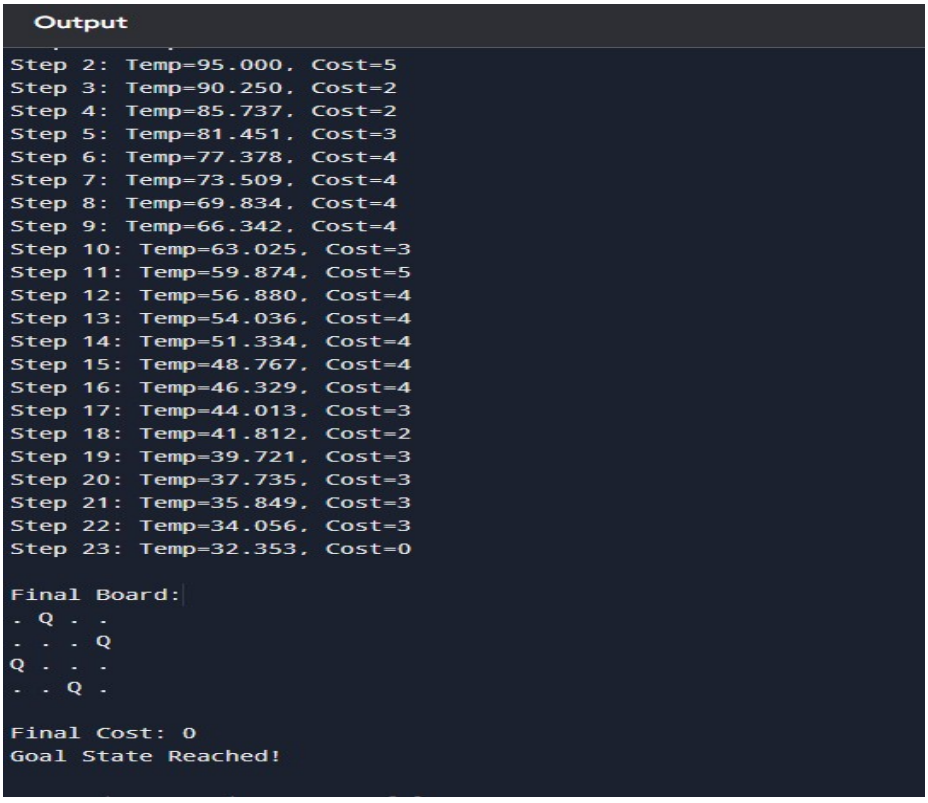


Code

```
import random
import time


def generate_initial_state(n=4):
    return [random.randint(0, n - 1) for _ in range(n)]


def calculate_conflicts(state):
    conflicts = 0
    n = len(state)
    for i in range(n):
        for j in range(i + 1, n):
            if state[i] == state[j]:
                conflicts += 1
            if abs(state[i] - state[j]) == abs(i - j):
                conflicts += 1
    return conflicts


def get_neighbors(state):
    neighbors = []
    n = len(state)
    for col in range(n):
        for row in range(n):
            if state[col] != row:
                neighbor = state.copy()
                neighbor[col] = row
                neighbors.append(neighbor)
    return neighbors


def print_board(state):
    n = len(state)
    for row in range(n):
        line = ""
        for col in range(n):
            if state[col] == row:
                line += "Q "
```

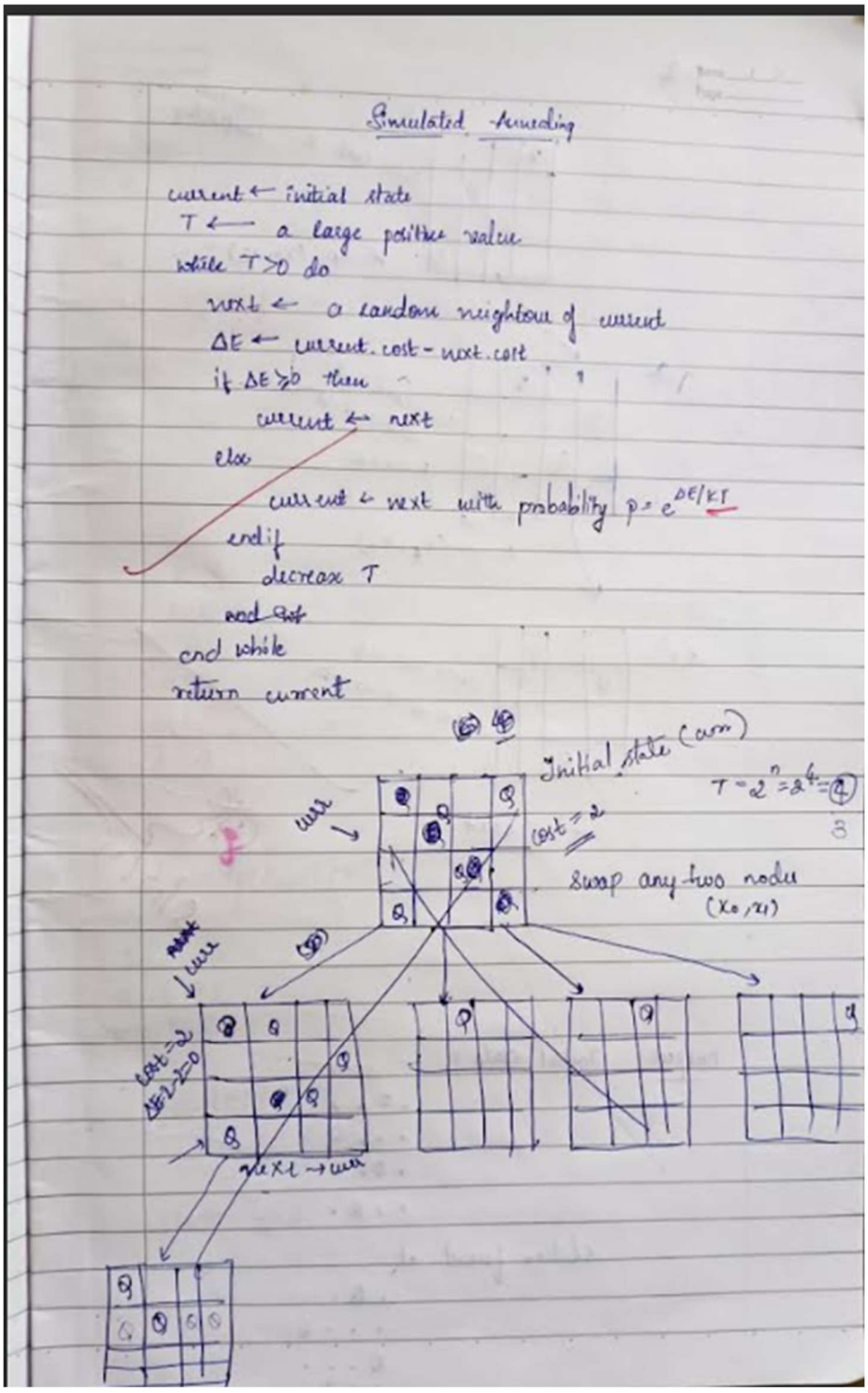
```
            else:
                line += ". "
        print(line)
    print()


def hill_climbing_with_steps(n=4, max_restarts=100):
    for restart in range(max_restarts):
        current = generate_initial_state(n)
        step = 0
        print(f"Restart #{restart + 1}: Initial state (Conflicts = {calculate_conflicts(current)})")
        print_board(current)

        while True:
            neighbors = get_neighbors(current)
            current_conflicts = calculate_conflicts(current)

            # Select best neighbor
            best_neighbor = min(neighbors, key=calculate_conflicts)
            best_conflicts = calculate_conflicts(best_neighbor)

            # If no improvement, break
            if best_conflicts >= current_conflicts:
                print(f"No improvement at step {step}. Restarting...\n")
                break

            # Move to better state
            current = best_neighbor
            step += 1
            print(f"Step {step}: Conflicts = {best_conflicts}")
            print_board(current)

            # Goal reached
            if best_conflicts == 0:
                print("Solution found!")
                print_board(current)
                return current

    print("No solution found after all restarts.")
    return None
```

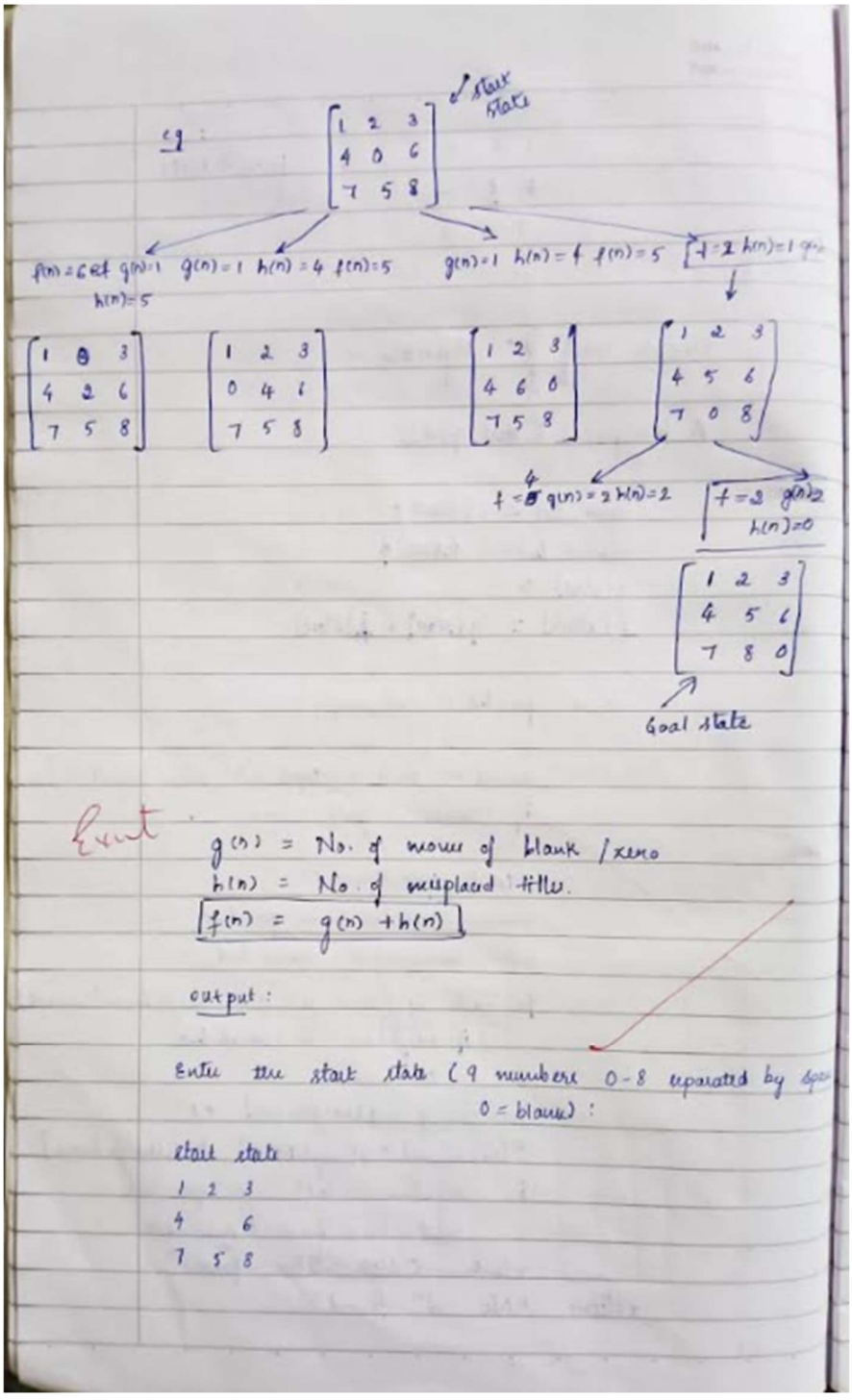
ScreenShot:


Output
Step 2: Temp=95.000, Cost=5
Step 3: Temp=90.250, Cost=2
Step 4: Temp=85.737, Cost=2
Step 5: Temp=81.451, Cost=3
Step 6: Temp=77.378, Cost=4
Step 7: Temp=73.509, Cost=4
Step 8: Temp=69.834, Cost=4
Step 9: Temp=66.342, Cost=4
Step 10: Temp=63.025, Cost=3
Step 11: Temp=59.874, Cost=5
Step 12: Temp=56.880, Cost=4
Step 13: Temp=54.036, Cost=4
Step 14: Temp=51.334, Cost=4
Step 15: Temp=48.767, Cost=4
Step 16: Temp=46.329, Cost=4
Step 17: Temp=44.013, Cost=3
Step 18: Temp=41.812, Cost=2
Step 19: Temp=39.721, Cost=3
Step 20: Temp=37.735, Cost=3
Step 21: Temp=35.849, Cost=3
Step 22: Temp=34.056, Cost=3
Step 23: Temp=32.353, Cost=0
Final Board:
. Q . .
. . . Q
Q . . .
. . Q .
Final Cost: 0
Goal State Reached!

Program 6:

Write a program to implement Simulated Annealing Algorithm :



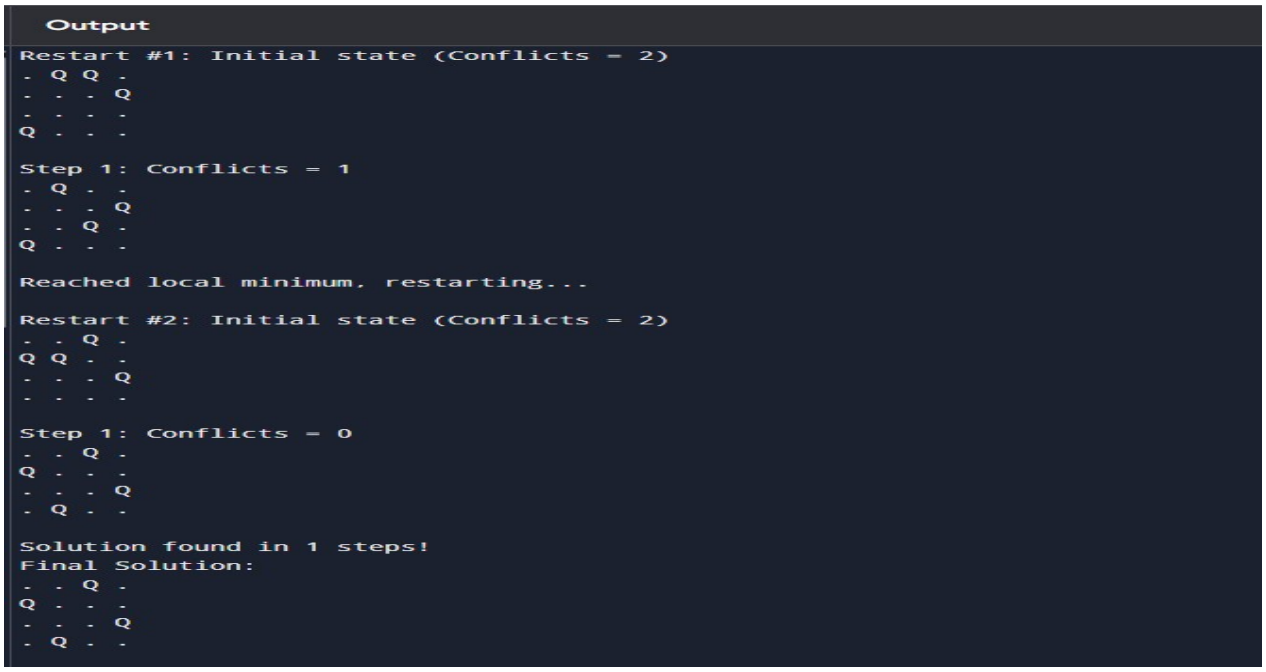


Code:

```
import random
import math
def print_board(board):
    n = len(board)
    for i in range(n):
```

```
        row = ["Q" if board[i] == j else "." for j in range(n)]
        print(" ".join(row))
    print()
def calculate_cost(board):
    """Heuristic: number of pairs of queens attacking each other"""
    n = len(board)
    cost = 0
    for i in range(n):
        for j in range(i + 1, n):
            if board[i] == board[j] or abs(board[i] - board[j]) == abs(i - j):
                cost += 1
    return cost
def random_neighbor(board):
    """Generate a random neighboring board by moving one queen"""
    n = len(board)
    neighbor = list(board)
    row = random.randint(0, n - 1)
    col = random.randint(0, n - 1)
    neighbor[row] = col
    return neighbor
def simulated_annealing(n, initial_temp=100, cooling_rate=0.95, stopping_temp=1):
    current_board = [random.randint(0, n - 1) for _ in range(n)]
    current_cost = calculate_cost(current_board)
    temperature = initial_temp
    step = 1
    print("Initial Board:")
    print_board(current_board)
    print(f"Initial Cost: {current_cost}\n")
    while temperature > stopping_temp and current_cost > 0:
        neighbor = random_neighbor(current_board)
        neighbor_cost = calculate_cost(neighbor)
        delta = neighbor_cost - current_cost
        if delta < 0 or random.random() < math.exp(-delta / temperature):
            current_board = neighbor
            current_cost = neighbor_cost
        print(f"Step {step}: Temp={temperature:.3f}, Cost={current_cost}")
        step += 1
        temperature *= cooling_rate
    print("\nFinal Board:")
    print_board(current_board)
    print(f"Final Cost: {current_cost}")
    if current_cost == 0:
        print("Goal State Reached!")
    else:
    print("Terminated before reaching goal.") simulated_annealing
```

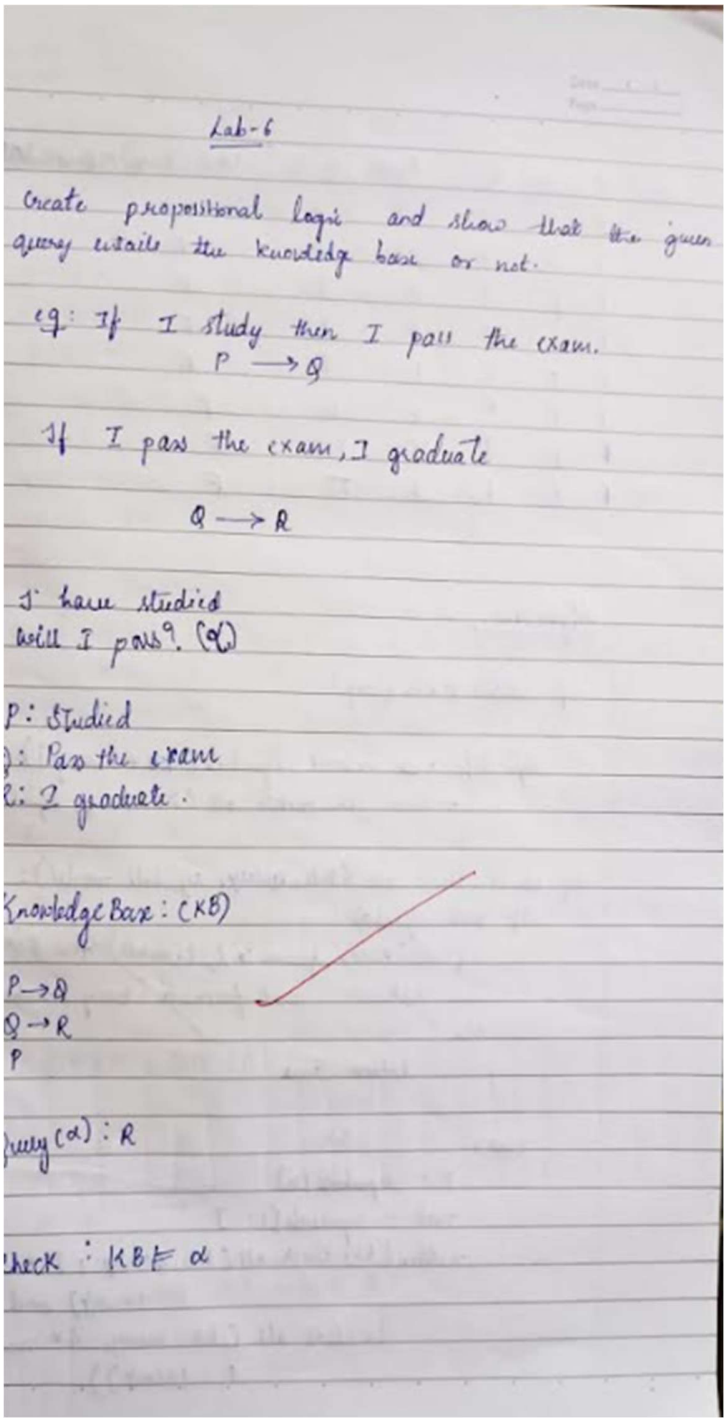
ScreenShot:


Output
Restart #1: Initial state (Conflicts = 2)
. Q Q .
. . . Q
. . . .
Q . . .
Step 1: Conflicts = 1
. Q . .
. . . Q
. . Q .
Q . . .
Reached local minimum, restarting...
Restart #2: Initial state (Conflicts = 2)
. . Q .
Q Q . .
. . . Q
. . . .
Step 1: Conflicts = 0
. . Q .
Q . . .
. . . Q
. Q . .
Solution found in 1 steps!
Final Solution:
. . Q .
Q . . .
. . . Q
. Q . .

Program 7:

Create a knowledge base using prepositional logic and show that the given query entails the knowledge base or not.

Algorithm:



Lab-6

Create propositional logic and show that the given query entails the knowledge base or not.

eg: If I study then I pass the exam.
P → Q

If I pass the exam, I graduate
Q → R

I have studied
will I pass? (α)

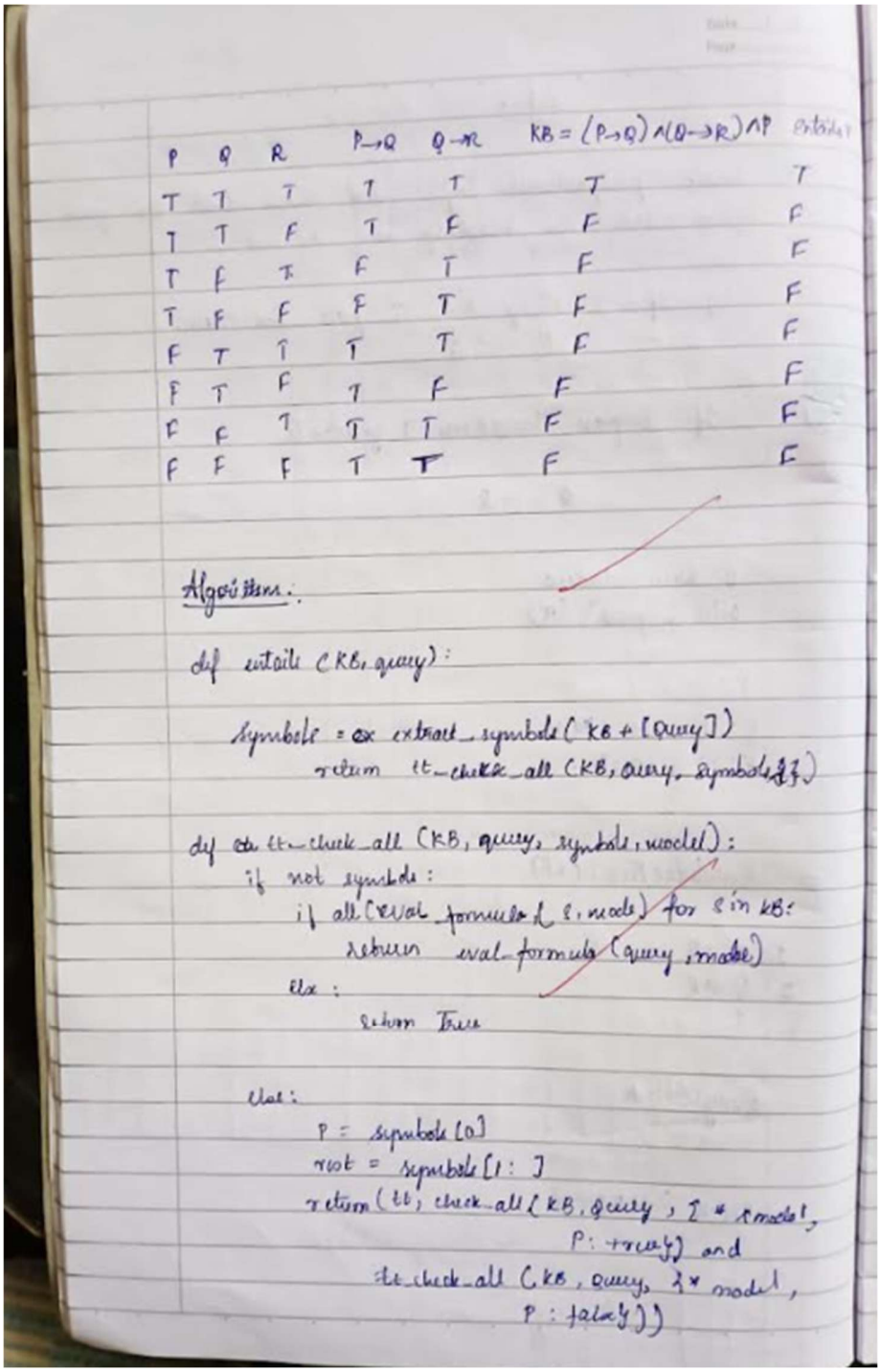
P: Studied
Q: Pass the exam
R: I graduate.

Knowledge Base: (KB)

P → Q
Q → R
P

Query (α): R

Check : KB ⊨ α



P	Q	R	P→Q	Q→R	KB = (P→Q)∧(Q→R)∧P	Entails
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	T	F	F
F	T	F	T	F	F	F
F	F	T	T	T	F	F
F	F	F	T	T	F	F

Algorithm:

```
def entails(KB, query):
    symbols = extract_symbols(KB + [query])
    return tt_check_all(KB, query, symbols, {})

def tt_check_all(KB, query, symbols, model):
    if not symbols:
        if all(eval_formula(s, model) for s in KB):
            return eval_formula(query, model)
        else:
            return True
    else:
        P = symbols[0]
        rest = symbols[1:]
        return (tt_check_all(KB, query, rest, {**model, P: True}) and
                tt_check_all(KB, query, rest, {**model, P: False}))
```

Consider a Knowledge base KB that contains the following propositional logic

$Q \rightarrow P$

$P \rightarrow \neg Q$

$Q \vee R$

i) Construct a TT that shows the truth value of each sentence in KB and indicate the models in which KB is true.

P	Q	R	$Q \rightarrow P$	$P \rightarrow \neg Q$	$Q \vee R$	KB entails (R)?
T	T	T	T	F	T	F
T	T	F	T	F	T	F
T	F	T	T	T	T	T (entails)
T	F	F	T	T	F	F
F	T	T	F	T	T	F
F	T	F	F	T	T	F
F	F	T	T	T	T	T (entails)
F	F	F	T	T	F	F

ii) Does KB entail (R) ?

P	Q	R	KB
T	F	T	T
F	F	T	T

So KB entails R.

(iii) Does KB entail $R \rightarrow P$

$R \rightarrow P$	KB
$T \rightarrow T$ (T)	T
$T \rightarrow F$ (F)	T

$\therefore$ Does not entail $R \rightarrow P$

(iv) Does KB entail $Q \rightarrow R$

$Q \rightarrow R$	KB
$(F \rightarrow T)$ T	T
$(F \rightarrow T)$ T	T

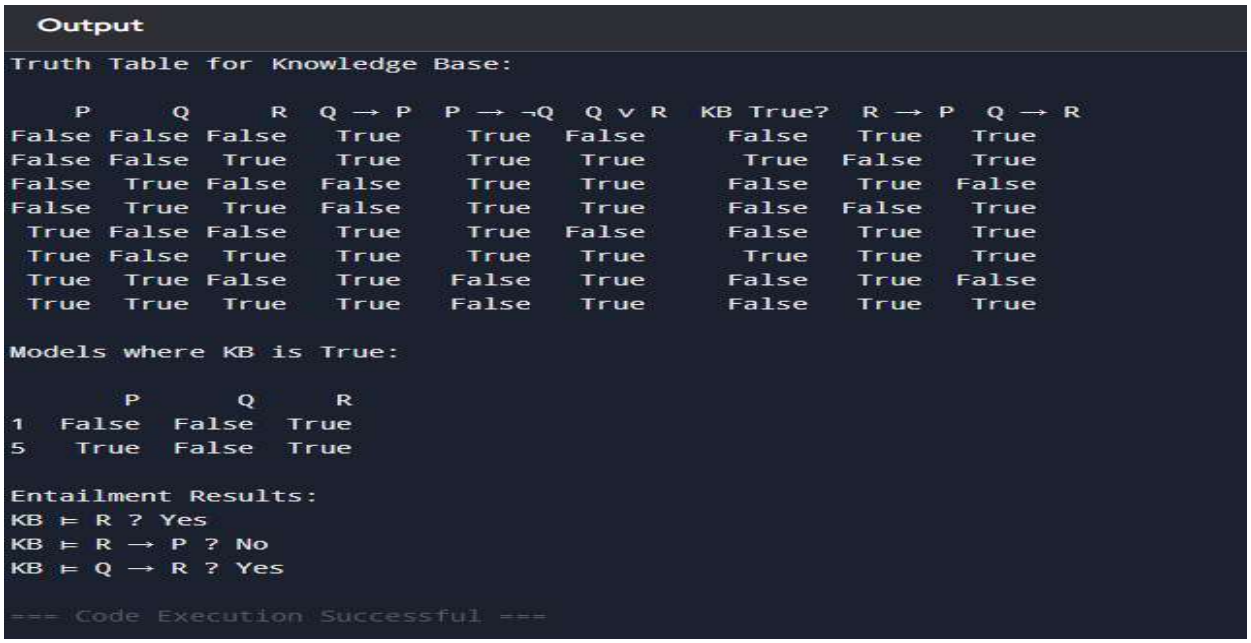
$\therefore$ Does entail $Q \rightarrow R$.

15/10/25

Code:

```
import itertools
import pandas as pd
variables = ['P', 'Q', 'R']
combinations = list(itertools.product([False, True], repeat=3))
rows = []
for (P, Q, R) in combinations:
    s1 = (not Q) or P
    s2 = (not P) or (not Q)  # P → ¬Q
    s3 = Q or R              # Q ∨ R
    KB = s1 and s2 and s3
    entail_R = R

    entail_R_imp_P = (not R) or P
    entail_Q_imp_R = (not Q) or R
    rows.append({
        'P': P, 'Q': Q, 'R': R,
        'Q → P': s1,
        'P → ¬Q': s2,
        'Q ∨ R': s3,
        'KB True?': KB,
        'R': entail_R,
        'R → P': entail_R_imp_P,
        'Q → R': entail_Q_imp_R
    })
df = pd.DataFrame(rows)
print("Truth Table for Knowledge Base:\n")
print(df.to_string(index=False))
models_true = df[df['KB True?'] == True]
print("\nModels where KB is True:\n")
print(models_true[['P', 'Q', 'R']])
def entails(column):
    """Check if KB entails the given statement."""
    return all(models_true[column])
print("\nEntailment Results:")
print(f"KB ⊨ R ? {'Yes' if entails('R') else 'No'}")
print(f"KB ⊨ R → P ? {'Yes' if entails('R → P') else 'No'}")
print(f"KB ⊨ Q → R ? {'Yes' if entails('Q → R') else 'No'}")
```

ScreenShot:

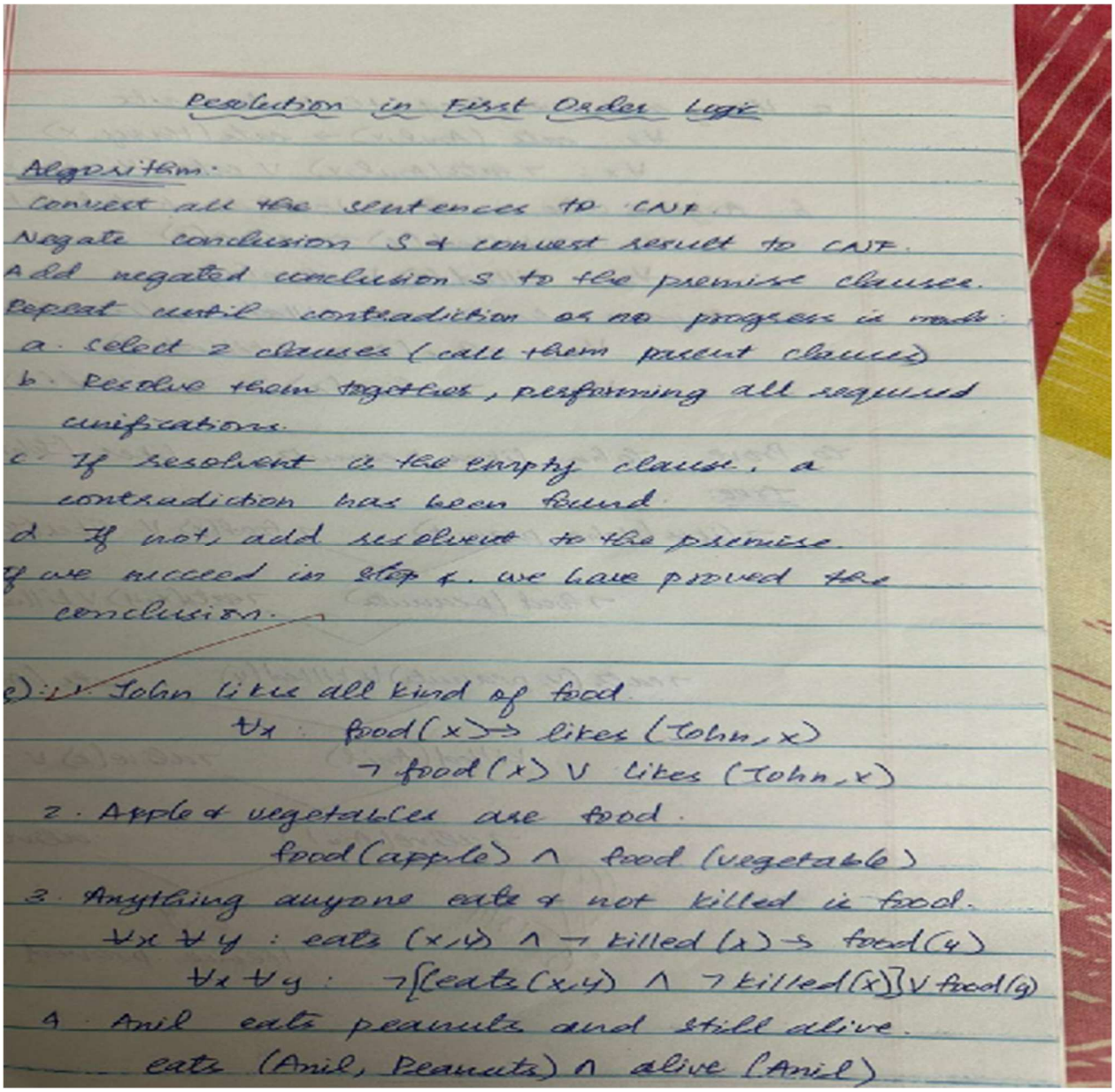


```
Output
Truth Table for Knowledge Base:

    P      Q      R   Q → P   P → ¬Q   Q ∨ R   KB True?   R → P   Q → R
False  False  False    True     True   False      False    True    True
False  False   True    True     True    True       True   False    True
False   True  False   False     True    True      False    True   False
False   True   True   False     True    True      False   False    True
 True  False  False    True     True   False      False    True    True
 True  False   True    True     True    True       True    True    True
 True   True  False    True    False    True      False    True   False
 True   True   True    True    False    True      False    True    True

Models where KB is True:

       P      Q      R
1  False  False   True
5   True  False   True

Entailment Results:
KB ⊨ R ? Yes
KB ⊨ R → P ? No
KB ⊨ Q → R ? Yes

=== Code Execution Successful ===
```

Program 8:

Create a knowledge base using prepositional logic and prove the given query using resolution.

Algorithm:



Resolution in First Order Logic

Algorithm:
Convert all the sentences to CNF.
Negate conclusion S & convert result to CNF.
Add negated conclusion S to the premise clauses.
Repeat until contradiction or no progress is made:
a. Select 2 clauses (call them parent clauses)
b. Resolve them together, performing all required unifications.
c. If resolvent is the empty clause, a contradiction has been found
d. If not, add resolvent to the premise.
If we succeed in step 4, we have proved the conclusion.

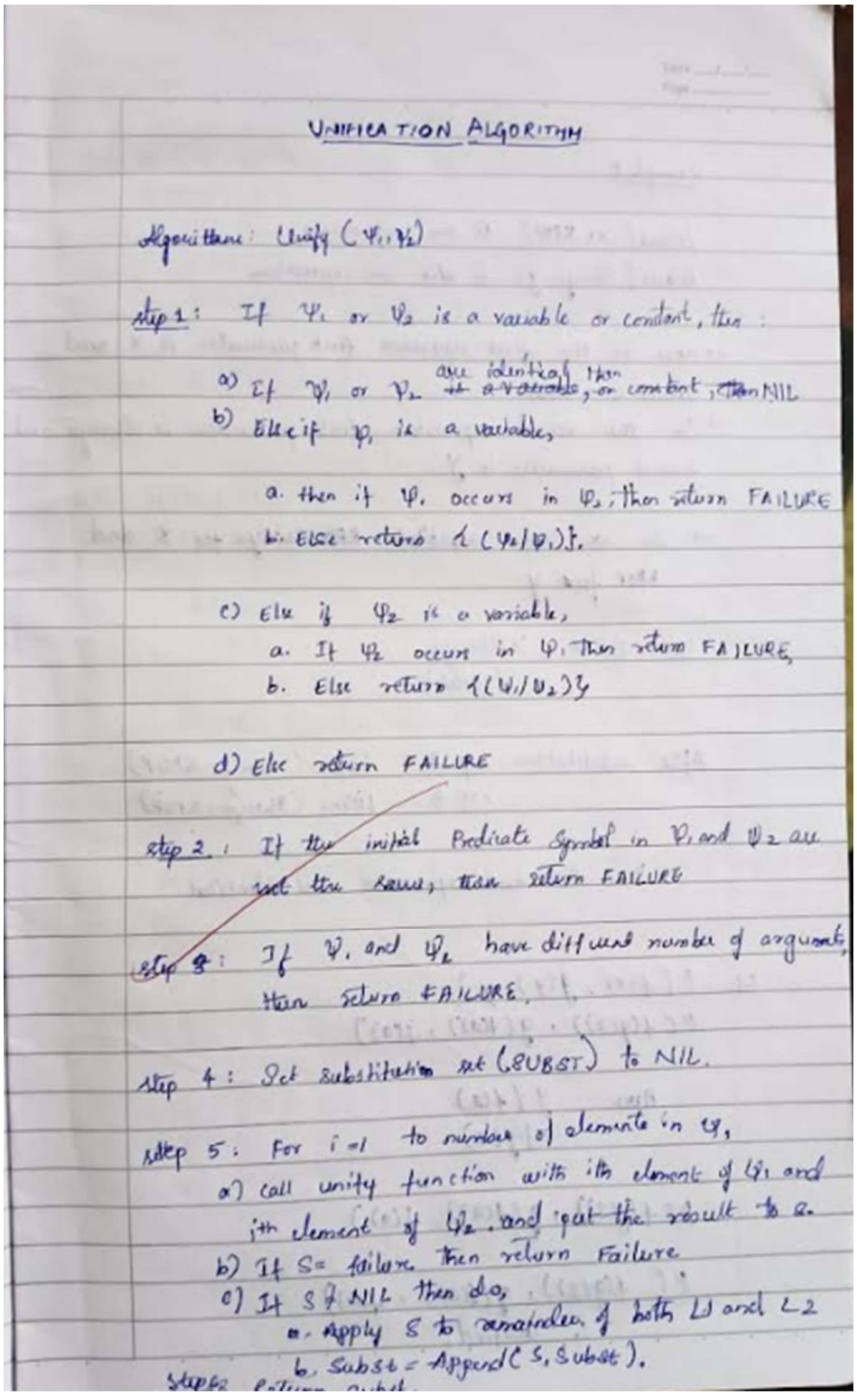
e): 1. John likes all kind of food.
$\forall x$: food(x) → likes(John, x)
$\neg$ food(x) $\vee$ likes(John, x)
2. Apple & vegetables are food.
food(apple) $\wedge$ food(vegetable)
3. Anything anyone eats & not killed is food.
$\forall x \forall y$: eats(x, y) $\wedge \neg$ killed(x) → food(y)
$\forall x \forall y$: $\neg$ [eats(x, y) $\wedge \neg$ killed(x)] $\vee$ food(y)
4. Anil eats peanuts and still alive.
eats(Anil, Peanuts) $\wedge$ alive(Anil)

```
import itertools
import pandas as pd

variables = ['P', 'Q', 'R']
combinations = list(itertools.product([False, True], repeat=3))

rows = []

for (P, Q, R) in combinations:
    s1 = (not Q) or P          # Q → P
    s2 = (not P) or (not Q)    # P → ¬Q
    s3 = Q or R                # Q ∨ R

    KB = s1 and s2 and s3

    entail_R = R
    entail_R_imp_P = (not R) or P      # R → P
    entail_Q_imp_R = (not Q) or R      # Q → R

    rows.append({
        'P': P,
        'Q': Q,
        'R': R,
        'Q → P': s1,
        'P → ¬Q': s2,
        'Q ∨ R': s3,
        'KB True?': KB,
        'R': entail_R,
        'R → P': entail_R_imp_P,
        'Q → R': entail_Q_imp_R
    })

df = pd.DataFrame(rows)

print("Truth Table for Knowledge Base:\n")
print(df.to_string(index=False))

models_true = df[df['KB True?'] == True]

print("\nModels where KB is True:\n")
```

```
print(models_true[['P', 'Q', 'R']])


def entails(column):
    """Check if KB entails the given statement."""
    return all(models_true[column])


print("\nEntailment Results:")
print(f"KB ⊨ R ? {'Yes' if entails('R') else 'No'}")
print(f"KB ⊨ R → P ? {'Yes' if entails('R → P') else 'No'}")
print(f"KB ⊨ Q → R ? {'Yes' if entails('Q → R') else 'No'}")
```

Program 9:

Implement unification in first order logic.
Algorithm:



UNIFICATION ALGORITHM

Algorithm: Unify (Ψ_1, Ψ_2)

step 1: If Ψ_1 or Ψ_2 is a variable or constant, then:

a) If Ψ_1 or Ψ_2 are identical then return NIL
b) Else if Ψ_1 is a variable,

a. then if Ψ_1 occurs in Ψ_2, then return FAILURE
b. Else return {(Ψ_2/Ψ_1)}.

c) Else if Ψ_2 is a variable,
a. If Ψ_2 occurs in Ψ_1 then return FAILURE,
b. Else return {(Ψ_1/Ψ_2)}

d) Else return FAILURE

step 2: If the initial Predicate Symbol in Ψ_1 and Ψ_2 are not the same, then return FAILURE

step 3: If Ψ_1 and Ψ_2 have different number of arguments, then return FAILURE.

step 4: Set substitution set (SUBST) to NIL.

step 5: For i=1 to number of elements in Ψ_1,
a) call unify function with ith element of Ψ_1 and ith element of Ψ_2, and put the result to S.
b) If S= failure then return Failure
c) If S ≠ NIL then do,
a. Apply S to remainder of both L1 and L2
b. Subst = Append(S, Subst).

step 6: return Subst

Example:

Listens(x, KPOP) is an expression
Listens(Shreya, y) is also an expression

→ now in the first expression first parameter is x and second is KPOP
→ In the second expression first parameter is Shreya and second parameter is y.
→ So we can substitute ~~KPOP~~ Shreya for x and KPOP for y

so $\theta = \{ x/Shreya, y/KPOP \}$

After substitution exp A : Listens(Shreya, KPOP)
exp B : Listens (Shreya, KPOP)

Thus the two expressions are identical

Q1. P(f(x), g(y), y)
P(f(g(z)), g(f(a)), f(a))

Here $y / f(a)$
$x / g(z)$

P(f(g(z)), g(f(a)), f(a))

P(f(g(z)), g(f(a)), f(a))
∴ unified.

Q2. $Q(x, f(x))$
$Q(f(y), y)$

$x \mid f(y)$ x appears in both and y appears
$y \mid f(x)$ in both the expression then it is not unifyable.

$\therefore Q(f(y), f(x))$

Q3. $H(x, g(z))$
$H(g(y), g(g(z)))$

$x \mid g(y)$
~~$x \mid g(z)$~~

~~x can't take two values~~

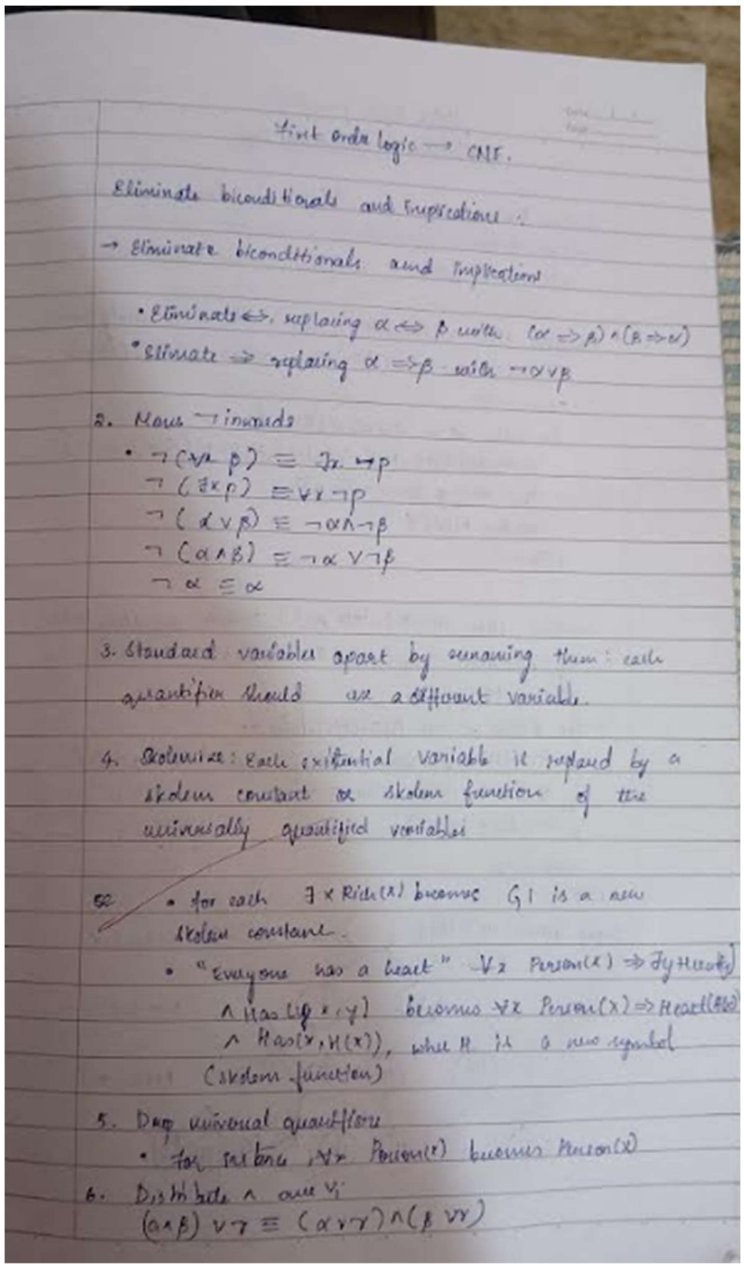
$y \mid z$

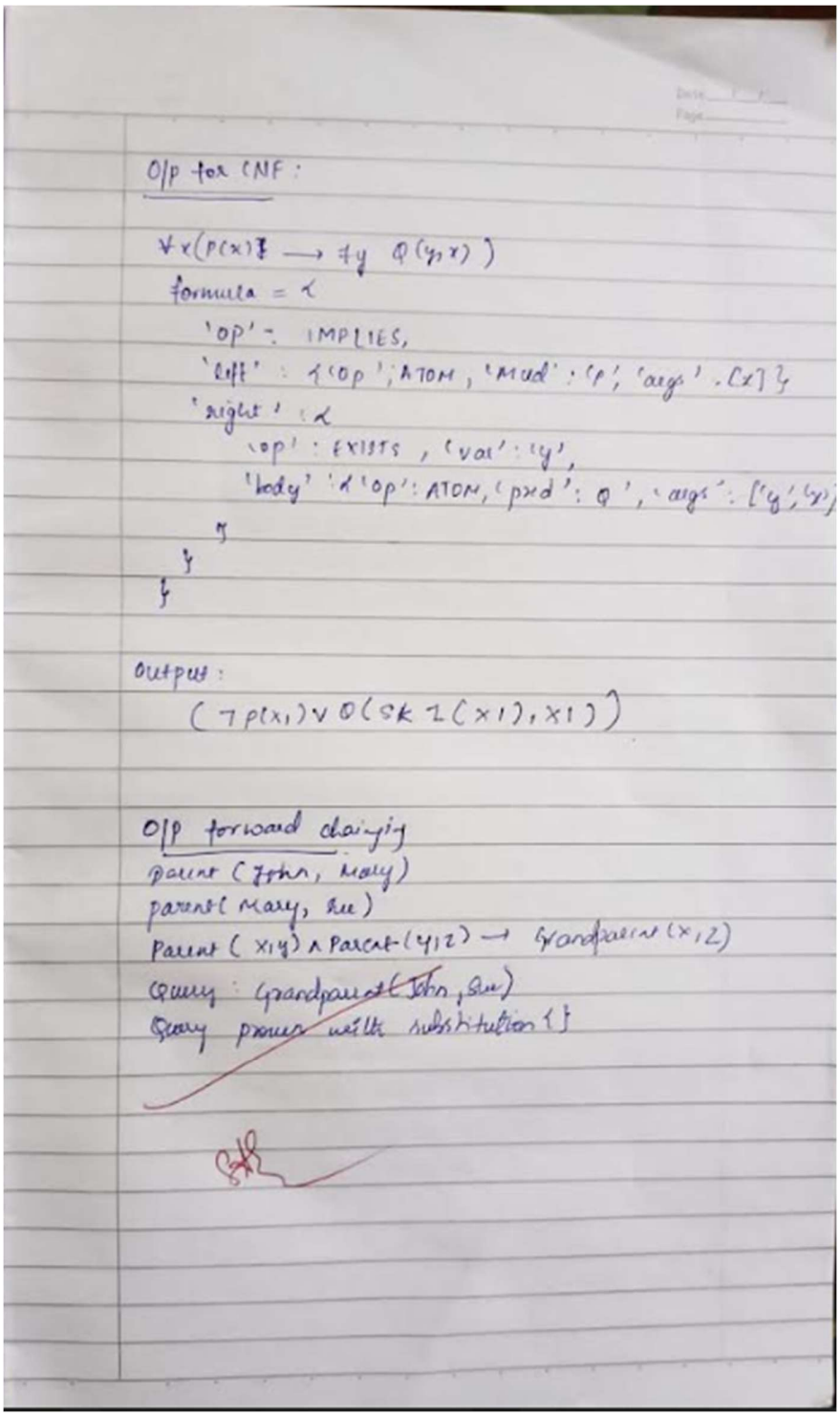
So expressions are: $H(g(y), g(g(z)))$ and $H(g(y), g(g(z)))$

8.3
29/10/23

Program 10:

Convert a given first order logic statement into Conjunctive Normal Form (CNF).
Algorithm:





O/p for CNF :

$\forall x(p(x) \rightarrow \exists y\ Q(y,x))$

formula = {

'op': IMPLIES,

'left': {'op': ATOM, 'pred': 'p', 'args': [x]},

'right': {

'op': EXISTS, 'var': 'y',

'body': {'op': ATOM, 'pred': 'Q', 'args': ['y', 'x']}

}

}

Output :

$(\neg p(x_1) \vee Q(Sk_1(x_1), x_1))$

O/p forward chaining

parent (John, Mary)

parent (Mary, Sue)

Parent (x, y) ∧ Parent (y, z) → Grandparent (x, z)

Query : Grandparent (John, Sue)

Query proven with substitution {}

import itertools

```
# ------------------------------------------------------------
# Use ASCII strings as operator names
# ------------------------------------------------------------
FORALL = 'FORALL'       # ∀
EXISTS = 'EXISTS'       # ∃
NOT = 'NOT'             # ¬
AND = 'AND'             # ∧
OR = 'OR'               # ∨
IMPLIES = 'IMPLIES'     # →
IFF = 'IFF'             # ↔
ATOM = 'ATOM'

# ------------------------------------------------------------
# Unicode symbols for pretty printing only
# ------------------------------------------------------------
unicode_symbols = {
    FORALL: '∀',
    EXISTS: '∃',
    NOT: '¬',
    AND: '∧',
    OR: '∨',
    IMPLIES: '→',
    IFF: '↔'
}

# ------------------------------------------------------------
# Utility counters for skolem and variables
# ------------------------------------------------------------
_skolem_counter = itertools.count(1)
_var_counter = itertools.count(1)

def new_skolem_name():
    return f"SK{next(_skolem_counter)}"

def new_var_name():
    return f"X{next(_var_counter)}"

# ------------------------------------------------------------
# STEP 1: Eliminate ↔ and →
# ------------------------------------------------------------
def eliminate_implications(F):
    if F is None:
        return None

    op = F['op']
```

```
    if op == ATOM:
        return F
    elif op == IMPLIES:
        # (α → β) ≡ (¬α ∨ β)
        return {'op': OR,
                'left': {'op': NOT, 'body': eliminate_implications(F['left'])},
                'right': eliminate_implications(F['right'])}
    elif op == IFF:
        # (α ↔ β) ≡ (α → β) ∧ (β → α)
        p = eliminate_implications(F['left'])
        q = eliminate_implications(F['right'])
        return {'op': AND,
                'left': {'op': OR, 'left': {'op': NOT, 'body': p}, 'right': q},
                'right': {'op': OR, 'left': {'op': NOT, 'body': q}, 'right': p}}
    elif op == NOT:
        return {'op': NOT, 'body': eliminate_implications(F['body'])}
    elif op in (AND, OR):
        return {'op': op,
                'left': eliminate_implications(F['left']),
                'right': eliminate_implications(F['right'])}
    elif op in (FORALL, EXISTS):
        return {'op': op, 'var': F['var'], 'body': eliminate_implications(F['body'])}
    return F

# ---------------------------------------------------------
# STEP 2: Move negations inward
# ---------------------------------------------------------
def move_negations_inward(F):
    if F['op'] == ATOM:
        return F
    elif F['op'] == NOT:
        sub = F['body']
        if sub['op'] == ATOM:
            return F
        elif sub['op'] == NOT:
            return move_negations_inward(sub['body'])
        elif sub['op'] == AND:
            return {'op': OR,
                    'left': move_negations_inward({'op': NOT, 'body': sub['left']}),
                    'right': move_negations_inward({'op': NOT, 'body': sub['right']})}
        elif sub['op'] == OR:
            return {'op': AND,
                    'left': move_negations_inward({'op': NOT, 'body': sub['left']}),
                    'right': move_negations_inward({'op': NOT, 'body': sub['right']})}
        elif sub['op'] == FORALL:
            return {'op': EXISTS, 'var': sub['var'],
                    'body': move_negations_inward({'op': NOT, 'body': sub['body']})}
```

```
        elif sub['op'] == EXISTS:
            return {'op': FORALL, 'var': sub['var'],
                    'body': move_negations_inward({'op': NOT, 'body': sub['body']})}
    elif F['op'] in (AND, OR):
        return {'op': F['op'],
                'left': move_negations_inward(F['left']),
                'right': move_negations_inward(F['right'])}
    elif F['op'] in (FORALL, EXISTS):
        return {**F, 'body': move_negations_inward(F['body'])}
    return F

# ---------------------------------------------------------
# STEP 3: Standardize variables
# ---------------------------------------------------------
def standardize_variables(F, mapping=None):
    if mapping is None:
        mapping = {}
    if F['op'] == ATOM:
        args = [mapping.get(a, a) for a in F['args']]
        return {'op': ATOM, 'pred': F['pred'], 'args': args}
    elif F['op'] in (AND, OR):
        return {'op': F['op'],
                'left': standardize_variables(F['left'], mapping),
                'right': standardize_variables(F['right'], mapping)}
    elif F['op'] == NOT:
        return {'op': NOT, 'body': standardize_variables(F['body'], mapping)}
    elif F['op'] in (FORALL, EXISTS):
        new_var = new_var_name()
        mapping2 = mapping.copy()
        mapping2[F['var']] = new_var
        return {'op': F['op'], 'var': new_var, 'body': standardize_variables(F['body'], mapping2)}
    return F

# ---------------------------------------------------------
# STEP 4: Skolemize (remove ∃)
# ---------------------------------------------------------
def skolemize(F, scope_vars=None):
    if scope_vars is None:
        scope_vars = []
    if F['op'] == ATOM:
        return F
    elif F['op'] == FORALL:
        return {'op': FORALL, 'var': F['var'], 'body': skolemize(F['body'], scope_vars + [F['var']])}
    elif F['op'] == EXISTS:
        skolem_name = new_skolem_name()
        skolem_term = skolem_name if not scope_vars else f"{skolem_name}({','.join(scope_vars)})"
        return skolemize(substitute(F['body'], F['var'], skolem_term), scope_vars)
```

```
    elif F['op'] in (AND, OR):
        return {'op': F['op'],
                'left': skolemize(F['left'], scope_vars),
                'right': skolemize(F['right'], scope_vars)}
    elif F['op'] == NOT:
        return {'op': NOT, 'body': skolemize(F['body'], scope_vars)}
    return F

def substitute(F, var, term):
    if F['op'] == ATOM:
        new_args = [term if a == var else a for a in F['args']]
        return {'op': ATOM, 'pred': F['pred'], 'args': new_args}
    elif F['op'] in (AND, OR):
        return {'op': F['op'],
                'left': substitute(F['left'], var, term),
                'right': substitute(F['right'], var, term)}
    elif F['op'] == NOT:
        return {'op': NOT, 'body': substitute(F['body'], var, term)}
    elif F['op'] in (FORALL, EXISTS):
        if F['var'] == var:
            return F
        return {**F, 'body': substitute(F['body'], var, term)}
    return F

# ---------------------------------------------------------
# STEP 5: Drop ∀ quantifiers
# ---------------------------------------------------------
def drop_universal_quantifiers(F):
    if F['op'] == FORALL:
        return drop_universal_quantifiers(F['body'])
    elif F['op'] in (AND, OR):
        return {'op': F['op'],
                'left': drop_universal_quantifiers(F['left']),
                'right': drop_universal_quantifiers(F['right'])}
    elif F['op'] == NOT:
        return {'op': NOT, 'body': drop_universal_quantifiers(F['body'])}
    return F

# ---------------------------------------------------------
# STEP 6: Distribute ∨ over ∧
# ---------------------------------------------------------
def distribute_or_over_and(F):
    if F['op'] == OR:
        A = distribute_or_over_and(F['left'])
        B = distribute_or_over_and(F['right'])
        if A['op'] == AND:
            return {'op': AND,
```

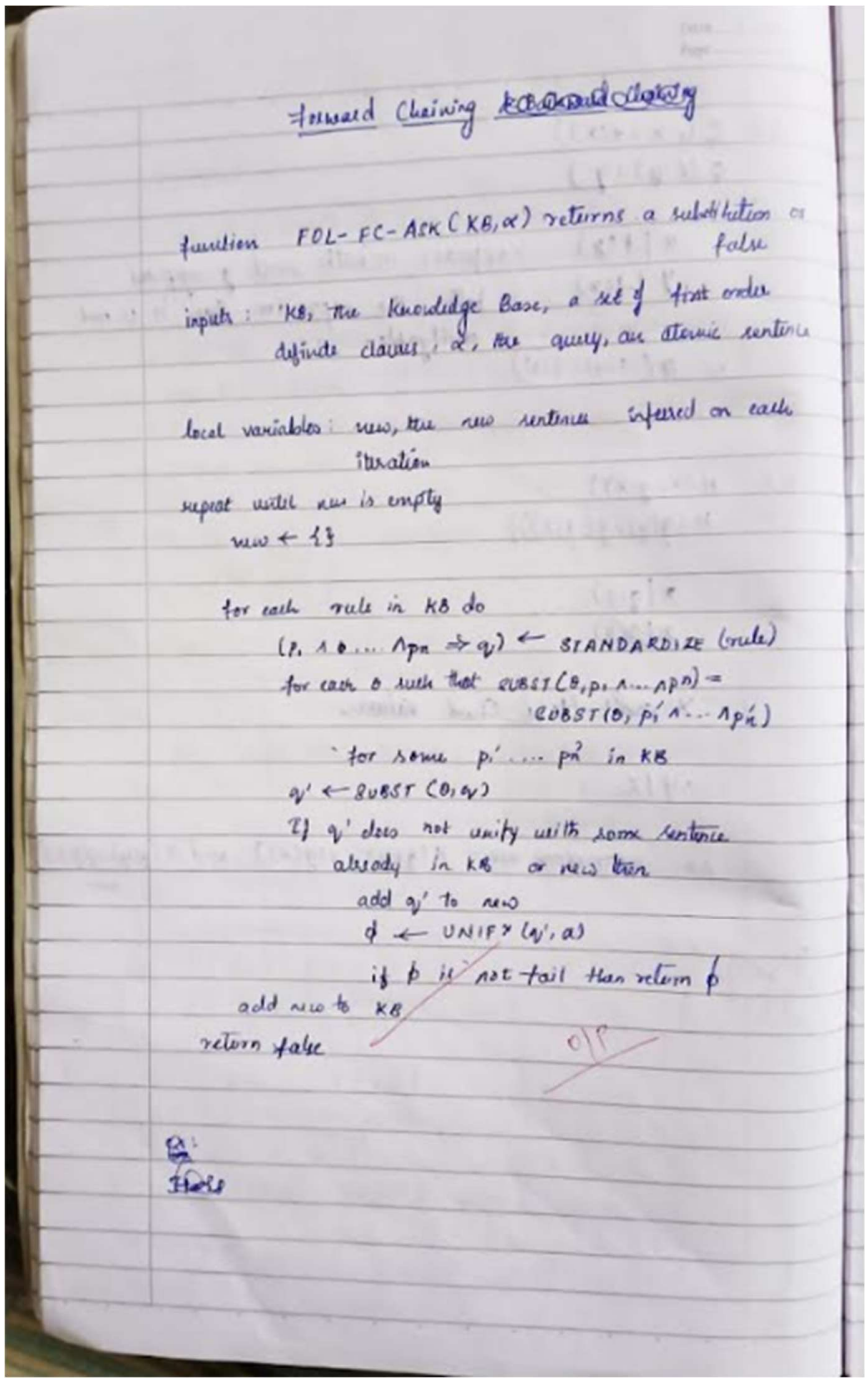
```
                'left': distribute_or_over_and({'op': OR, 'left': A['left'], 'right': B}),
                'right': distribute_or_over_and({'op': OR, 'left': A['right'], 'right': B})}
        elif B['op'] == AND:
            return {'op': AND,
                'left': distribute_or_over_and({'op': OR, 'left': A, 'right': B['left']}),
                'right': distribute_or_over_and({'op': OR, 'left': A, 'right': B['right']})}
        else:
            return {'op': OR, 'left': A, 'right': B}
    elif F['op'] == AND:
        return {'op': AND,
            'left': distribute_or_over_and(F['left']),
            'right': distribute_or_over_and(F['right'])}
    else:
        return F

# ---------------------------------------------------------
# PRETTY PRINTING (SYMBOLIC OUTPUT)
# ---------------------------------------------------------
def pretty(F):
    """Convert CNF structure to readable symbolic formula."""
    op = F['op']
    if op == ATOM:
        args = ",".join(F['args'])
        return f"{F['pred']}({args})"
    elif op == NOT:
        return f"¬{pretty(F['body'])}"
    elif op in (AND, OR):
        return f"({pretty(F['left'])} {unicode_symbols[op]} {pretty(F['right'])})"
    return str(F)

# ---------------------------------------------------------
# MAIN WRAPPER
# ---------------------------------------------------------
def to_CNF(F):
    F1 = eliminate_implications(F)
    F2 = move_negations_inward(F1)
    F3 = standardize_variables(F2)
    F4 = skolemize(F3)
    F5 = drop_universal_quantifiers(F4)
    F6 = distribute_or_over_and(F5)
    return F6

# ---------------------------------------------------------
# EXAMPLE
# ---------------------------------------------------------
if __name__ == "__main__":
    # ∀x ( P(x) → ∃y Q(y,x) )
```

```
formula = {
   'op': FORALL, 'var': 'x',
   'body': {
      'op': IMPLIES,
      'left': {'op': ATOM, 'pred': 'P', 'args': ['x']},
      'right': {
         'op': EXISTS, 'var': 'y',
         'body': {'op': ATOM, 'pred': 'Q', 'args': ['y', 'x']}
      }
   }
}

cnf = to_CNF(formula)
print("CNF Result (dictionary form):")
print(cnf)
print("\nCNF Result (symbolic form):")
print(pretty(cnf))
```

```
CNF Result (dictionary form):
{'op': 'OR', 'left': {'op': 'NOT', 'body': {'op': 'ATOM', 'pred': 'P', 'args': ['X1']}}, 'right': {'op': 'ATOM', 'pred': 'Q', 'args': ['SK1(X1)', 'X1']}}

CNF Result (symbolic form):
(¬P(X1) ∨ Q(SK1(X1),X1))
```

Program 11:

Create a knowledge base consisting of first order logic statements and prove the given query using forward reasoning.
Algorithm:



Forward Chaining

function FOL-FC-ASK(KB, α) returns a substitution or false

inputs: KB, the knowledge base, a set of first order definite clauses
α, the query, an atomic sentence

local variables: new, the new sentences inferred on each iteration

repeat until new is empty
new ← {}

for each rule in KB do
(p1 ∧ ... ∧ pn ⇒ q) ← STANDARDIZE(rule)
for each θ such that SUBST(θ, p1 ∧ ... ∧ pn) = SUBST(θ, p1' ∧ ... ∧ pn')
for some p1' ... pn' in KB
q' ← SUBST(θ, q)
If q' does not unify with some sentence already in KB or new then
add q' to new
ϕ ← UNIFY(q', α)
if ϕ is not fail then return ϕ
add new to KB
return false

```
import re

def isVariable(x):
    """
    Heuristically checks if a string is a variable (single, lowercase, alphabetic).
    """
    return len(x) == 1 and x.islower() and x.isalpha()

def getAttributes(string):
    """
    Extracts the parenthesized attribute strings. (e.g., 'P(a,b)' -> ['(a,b)'])
    Note: This uses simple regex and will struggle with nested functions.
    """
    expr = r'\([^)]+\)'
    matches = re.findall(expr, string)
    return matches

def getPredicates(string):
    """
    Extracts the predicate symbol. (e.g., 'P(a,b)' -> ['P'])
    """
    expr = r'([a-z~]+)\([^&|]+\)'
    return re.findall(expr, string)

class Fact:
    """
    Represents a single fact in the KB (e.g., 'Mother(a,b)').
    """
    def __init__(self, expression):
```

```
    self.expression = expression
    predicate, params = self.splitExpression(expression)
    self.predicate = predicate
    self.params = params
    # result is true if the fact contains at least one constant
    self.result = any(self.getConstants())

def splitExpression(self, expression):
    """
    Splits the expression into predicate and parameters.
    """
    predicate = getPredicates(expression)[0]
    params = getAttributes(expression)[0].strip('()').split(',')
    return [predicate, params]

def getResult(self):
    """
    Returns True if the fact has at least one constant.
    (Usage seems specific to this simplified inference process).
    """
    return self.result

def getConstants(self):
    """
    Returns a list of constants (or None for variables).
    """
    return [None if isVariable(c) else c for c in self.params]

def getVariables(self):
```

```
        """
        Returns a list of variables (or None for constants).
        """
        return [v if isVariable(v) else None for v in self.params]

    def substitute(self, constants):
        """
        Applies a substitution (constants) to create a new grounded Fact.
        (Seems unused in the evaluate method, but provided here.)
        """
        c = constants.copy()
        f = f"{self.predicate}({','.join([constants.pop(0) if isVariable(p) else p for p in self.params])})"
        return Fact(f)

class Implication:
    """
    Represents an implication rule (e.g., 'Mother(x,y)&Father(y,z) => Grandparent(x,z)').
    """
    def __init__(self, expression):
        self.expression = expression
        l = expression.split('=>')
        # LHS is a list of Facts connected by '&'
        self.lhs = [Fact(f) for f in l[0].split('&')]
        self.rhs = Fact(l[1])

    def evaluate(self, facts):
        """
        Attempts to apply the rule (forward chaining) using existing facts.
        This simplified version finds constants by position matching.
```

```
        """
        constants = {}
        new_lhs = [] # Facts that successfully matched LHS predicates

        for fact in facts:
            for val in self.lhs:
                if val.predicate == fact.predicate:
                    # Collect constants based on variable positions
                    for i, v in enumerate(val.getVariables()):
                        if v:
                            constants[v] = fact.getConstants()[i]
                    new_lhs.append(fact)

        # Check if we matched all LHS facts and if all matched facts are "true" (have constants)
        if len(new_lhs) == len(self.lhs) and all([f.getResult() for f in new_lhs]):
            # Construct the RHS fact by applying the collected constants
            predicate, attributes = getPredicates(self.rhs.expression)[0],
str(getAttributes(self.rhs.expression)[0])
            for key in constants:
                if constants[key]:
                    # Simple string replacement for substitution
                    attributes = attributes.replace(key, constants[key])

            expr = f'{predicate}{attributes}'
            return Fact(expr)

        return None

class KB:
    """
```

```
The Knowledge Base containing facts and implications.
"""
def __init__(self):
    self.facts = set()
    self.implications = set()

def tell(self, e):
    """
    Adds a new fact or implication to the KB and performs inference.
    """
    if '=>' in e:
        self.implications.add(Implication(e))
    else:
        self.facts.add(Fact(e))

    # Run forward chaining (simple form: only one pass)
    for i in self.implications:
        res = i.evaluate(self.facts)
        if res:
            self.facts.add(res)

def ask(self, e):
    """
    Searches the current facts for facts matching the query's predicate.
    """
    facts = set([f.expression for f in self.facts])
    i = 1
    print(f'Querying {e}:')
    for f in facts:
```

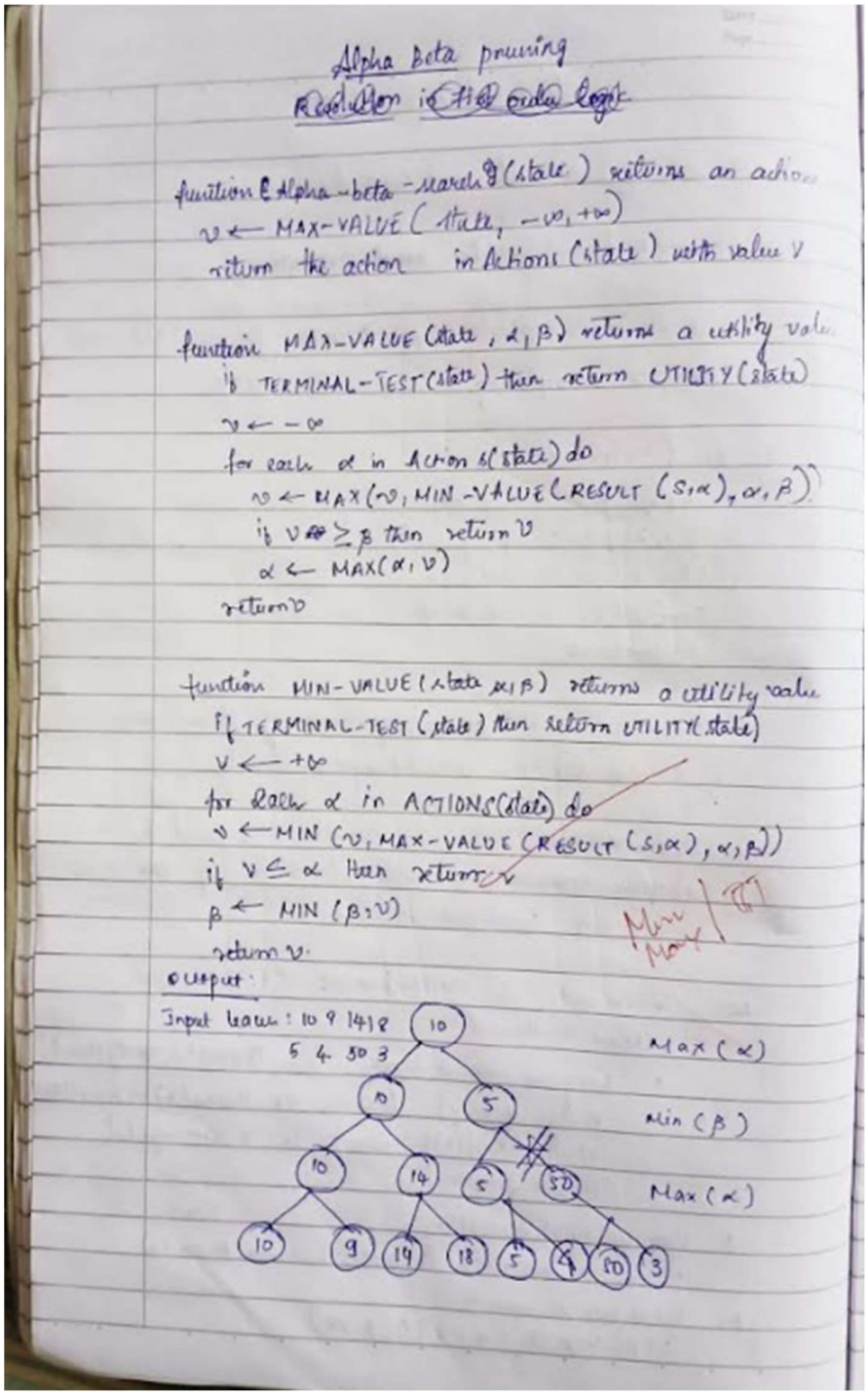
```
            if Fact(f).predicate == Fact(e).predicate:
                print(f'\t{i}. {f}')
                i += 1

    def display(self):
        """
        Prints all unique facts currently in the KB.
        """
        print("All facts: ")
        for i, f in enumerate(set([f.expression for f in self.facts])):
            print(f'\t{i+1}. {f}')

def main():
    """
    Main execution function to run the KB system.
    """
    kb = KB()
    print("Enter the number of FOL expressions present in KB:")
    # Example input: 2
    n = int(input())

    print("Enter the expressions:")
    # Example inputs:
    # Mother(ANN,BOB)
    # Mother(x,y)&Father(y,z) => Grandparent(x,z)
    for i in range(n):
        fact = input()
        kb.tell(fact)
```

```
        print("Enter the query:")
        # Example input: Grandparent(ANN,z)
        query = input()

        kb.ask(query)
        kb.display()

    if __name__ == "__main__":
main()
```

```
Enter the number of FOL expressions present in KB:
3
Enter the expressions:
Mother(ANN,BOB)
Father(BOB,CHARLIE)
Mother(x,y)&Father(y,z) => Grandparent(x,z)
Enter the query:
Grandparent(ANN,z)
Querying Grandparent(ANN,z):
    1. randparent(ANN,CHARLIE)
All facts:
    1. Mother(ANN,BOB)
    2. randparent(ANN,CHARLIE)
    3. Father(BOB,CHARLIE)
```

Program 12:

Implement Alpha-Beta Pruning.
Algorithm:



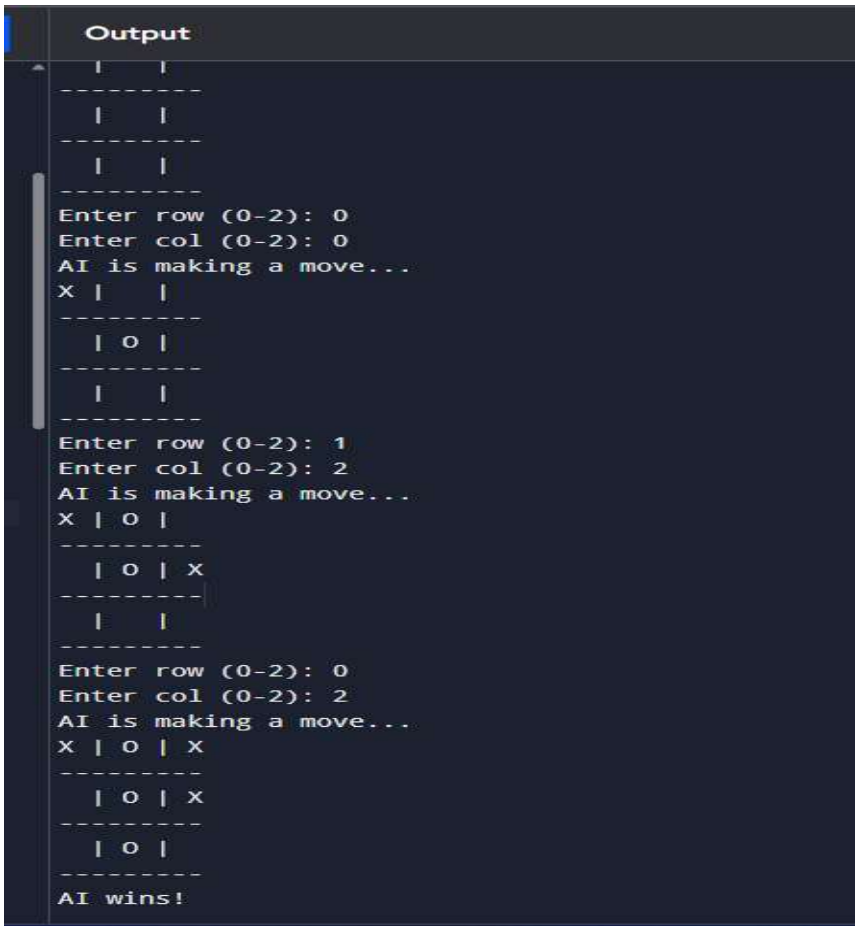
Code:

```
import math
PLAYER = "X"  # Human
AI = "O"      # Computer
def print_board(board):
    for row in board:
        print(" | ".join(row))
        print("-" * 9)
def available_moves(board):
    """Return list of available (row, col) moves."""
    moves = []

    for i in range(3):
        for j in range(3):
            if board[i][j] == " ":
                moves.append((i, j))
    return moves
def check_winner(board):
    """Return 'X' if X wins, 'O' if O wins, or None otherwise."""
    for i in range(3):
        if board[i][0] == board[i][1] == board[i][2] != " ":
            return board[i][0]
        if board[0][i] == board[1][i] == board[2][i] != " ":
            return board[0][i]
    if board[0][0] == board[1][1] == board[2][2] != " ":
        return board[0][0]
    if board[0][2] == board[1][1] == board[2][0] != " ":
        return board[0][2]
    return None
def is_full(board):
    return all(cell != " " for row in board for cell in row)
def minimax(board, depth, is_maximizing):
    winner = check_winner(board)
    if winner == AI:
        return 1
    elif winner == PLAYER:
        return -1
    elif is_full(board):
        return 0
    if is_maximizing:
        best_score = -math.inf
        for (i, j) in available_moves(board):
            board[i][j] = AI
            score = minimax(board, depth + 1, False)
            board[i][j] = " "
            best_score = max(score, best_score)
```

```
            return best_score
        else:
            best_score = math.inf
            for (i, j) in available_moves(board):
                board[i][j] = PLAYER
                score = minimax(board, depth + 1, True)
                board[i][j] = " "
                best_score = min(score, best_score)
            return best_score
    def best_move(board):
        """Find the best move for the AI."""
        best_score = -math.inf
        move = None

        for (i, j) in available_moves(board):
            board[i][j] = AI
            score = minimax(board, 0, False)
            board[i][j] = " "
            if score > best_score:
                best_score = score
                move = (i, j)
        return    move
    def play_game():
        board = [[" " for _ in range(3)] for _ in range(3)]
        print("Tic Tac Toe - You are X, AI is O")
        print_board(board)
        while True:
            row = int(input("Enter row (0-2): "))
            col = int(input("Enter col (0-2): "))
            if board[row][col] != " ":
                print("Cell taken, try again.")
                continue
            board[row][col] = PLAYER
            if check_winner(board) == PLAYER:
                print_board(board)
                print("You win!")
                break
            elif is_full(board):
                print_board(board)
                print("It's a draw!")
                break
            print("AI is making a move...")
            move = best_move(board)
if move:
```

```
            board[move[0]][move[1]] = AI
            print_board(board)
            if check_winner(board) == AI:
                print("AI wins!")
                break
            elif is_full(board):
                print("It's a draw!")
                break
    if __name__ == "__main__":
        play_game()
```

ScreenShot:



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